

Immigration Enforcement and Local Business Dynamics

Samyam Shrestha* Hugo Sant'Anna†

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Abstract

This paper examines the effects of interior immigration enforcement on local business dynamics. Exploiting the federally determined rollout of the Secure Communities program across U.S. counties, we use county-sector-year Census administrative data to show that enforcement reduces firm and establishment counts by suppressing establishment entry, with no significant effect on establishment exit. These effects are concentrated in high-immigrant counties and in immigrant-intensive, high-turnover, and non-tradable sectors, suggesting that immigration enforcement constrains business growth through reduced labor availability and local demand. Enforcement also reduces job creation, with the decline driven primarily by continuing establishments reducing expansion.

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*Department of Agricultural and Applied Economics, University of Georgia, 302 Conner Hall, Athens, GA 30602. Corresponding author.

†Collat School of Business, University of Alabama at Birmingham, CSB 261, 710 13th St S, Birmingham, AL 35233

1 Introduction

Immigration enforcement has expanded dramatically in the United States since the 1996 Illegal Immigration Reform and Immigrant Responsibility Act, which substantially increased the federal government’s capacity to identify, detain, and remove undocumented immigrants. Subsequent decades saw the expansion of interior enforcement programs that put these capacities into practice at scale. A growing empirical literature shows that enforcement reduces the undocumented population in affected areas and generates meaningful changes in local labor markets (Bohn et al., 2014; East et al., 2023; Orrenius and Zavodny, 2015; East and Velásquez, 2024). Yet this evidence is largely focused on worker outcomes such as employment and wages. We know much less about how enforcement-induced labor supply shocks affect local business activity, despite the fact that many industries depend heavily on low-skill immigrant labor and that business responses are the primary margin through which labor market disruptions translate into production, employment, and aggregate economic activity (Decker et al., 2014).

Local business dynamism can respond to such shocks in different ways, for instance by contracting on the entry and expansion margin rather than the exit and separation margin, with different economic implications even when aggregate employment effects look similar. Unlike layoffs or closures, a slowdown in business entry and expansion suppresses productive capacity in ways that are gradual and harder to observe in standard employment and wage data (Clementi and Palazzo, 2016; Decker et al., 2016). If enforcement primarily constrains entry and expansion rather than forcing incumbent establishments to exit, its economic costs accumulate through foregone growth rather than immediate dislocation, a distinction with direct implications for how policymakers evaluate the full consequences of interior enforcement.

In this paper, we estimate the effects of U.S. interior immigration enforcement on the number of local firms and establishments, and on the underlying establishment entry and exit margins that drive it, using variation in the rollout of Secure Communities (SC) across counties from 2008 to 2012. SC is a federal program that automatically checks the immigration status of individuals booked into local jails against federal databases, enabling the identification of unauthorized immigrants and potential removal. The program’s activation schedule was determined federally by ICE based on technical readiness and administrative capacity rather than local adoption decisions (Cox and Miles, 2013; East et al., 2023). This renders the timing of activation plausibly exogenous to local economic conditions, a claim we support empirically by showing that pre-treatment changes in county characteristics do not systematically predict the timing of SC adoption. We combine SC activation with U.S.

Census Bureau administrative data in a county-sector-year panel and estimate its effects using the staggered difference-in-differences estimator of [Sun and Abraham \(2021\)](#).

We find that SC activation reduces the number of establishments by 2.9% of the pre-treatment mean and the number of firms by 3.4%. This decline operates primarily through the entry margin: establishment entry falls by 9.5%, while exit is statistically indistinguishable from zero in the preferred specification, with event studies showing no evidence of pre-trends. This asymmetry is consistent with the broader literature finding that labor supply shocks operate primarily through the inflow margin rather than incumbent exit ([Dustmann et al., 2017](#); [Ortega and Verdugo, 2022](#)).

We trace these results through several additional margins of the analysis. First, SC activation reduces the Hispanic non-citizen population shares, with a similarly signed but statistically imprecise decline in the non-citizen share, confirming that enforcement reaches the population most exposed to the policy. Second, the decline in the number of firms, the number of establishments, and entry is concentrated in immigrant-intensive and high-turnover sectors and in high-immigrant counties, while exit remains statistically indistinguishable from zero in those groups. The decline in the number of firms and establishments is also larger in non-tradable than in tradable sectors, pointing to a possible role for local demand alongside labor supply.

Third, job creation falls, and the decline is concentrated among continuing establishments rather than at business births, indicating that enforcement restrains hiring at existing establishments more than it suppresses hiring by new entrants. Fourth, incorporated self-employment among non-citizens as well as citizens is statistically unchanged, as is the Census count of non-employer establishments, so the evidence points to a decline in the labor available to employer businesses, rather than a decline in immigrant entrepreneurship, as the principal explanation for the employer-business response.

Our results are robust to a wide range of alternative specifications. They hold under commuting-zone-by-year fixed effects and commuting-zone aggregation, and under region- and division-by-year fixed effects, which rule out differential regional recovery from the Great Recession as an alternative explanation. They are also stable under state-by-year fixed effects, [Conley \(1999\)](#) spatial standard errors, a spatial-noise placebo test, alternative clustering, trimming, and unweighted estimation. Three separate checks indicate that missing-data suppression does not drive these results. The [Borusyak et al. \(2024\)](#) imputation estimator confirms the direction and magnitude of the firm and establishment effects. Across all specifications, the number of firms, the number of establishments, and establishment entry decline, while exit remains statistically indistinguishable from zero.

This paper makes several contributions to the literature. First, it contributes to the literature on immigration enforcement by providing broad causal evidence on how a mandatory federal interior enforcement policy affects local business formation and employment dynamics. Prior work on immigration enforcement has focused primarily on worker outcomes: SC reduced employment among likely undocumented immigrants and lowered employment and wages for U.S.-born workers (East et al., 2023), generated broader equilibrium effects including higher prices for household services (East and Velásquez, 2024) and disruptions in immigrant-intensive local markets such as child care (Ali et al., 2024), and reduced labor market participation among immigrant women through network disruption (Bansak et al., 2025); more recently, heightened ICE enforcement during 2025 similarly reduced work participation among undocumented immigrants with no positive spillovers to native workers (Cox and East, 2026). Evidence on 287(g) programs, which permitted local law enforcement agents to perform some of the duties of ICE, documents sectoral employment declines (Bohn and Santillano, 2017), while firms exposed to intensified enforcement increase their reliance on guest worker programs as a substitute labor channel (Amuedo-Dorantes et al., 2021). This body of work establishes that enforcement disrupts labor markets and alters firm behavior. We complement it by using national county-sector business dynamics data to show how a mandatory federal enforcement shock propagates across business dynamism.

Second, the paper contributes to the literature on how local economies absorb labor supply shocks and how access to immigrant labor shapes firm behavior. A literature on business dynamism shows that labor force growth is a primary driver of firm entry rates, with a shrinking labor pool suppressing startup activity while leaving incumbent dynamics largely intact (Karahan et al., 2019; Hopenhayn et al., 2022), and that immigration inflows raise establishment counts, business survival, and employment (Olney, 2013; Orrenius et al., 2020; Mahajan et al., 2024). Immigrants also account for a disproportionate share of new firm formation and high-growth firms in the United States (Azoulay et al., 2022; Kerr and Kerr, 2020), and firm-level evidence shows that access to immigrant labor raises productivity and growth (Mitaritonna et al., 2017; Beerli et al., 2021), consistent with immigrant and native workers being complements (Peri and Sparber, 2009). This literature implies that restrictions on immigrant labor should suppress firm dynamics, a prediction consistent with evidence that E-Verify mandates reduce formal employment and firm activity (Ayromloo et al., 2020). Our findings extend this mechanism to mandatory federal interior enforcement, showing that reductions in immigrant labor supply lower firm and establishment counts by suppressing entry rather than inducing exit.

Third, this paper contributes to the literature on declining U.S. business dynamism. A large body of work documents secular declines in firm entry rates, job reallocation, and the

share of activity accounted for by young firms since the 1980s, with the decline becoming especially pronounced after 2000 (Haltiwanger et al., 2013; Decker et al., 2014; Davis and Haltiwanger, 2014; Decker et al., 2016). Karahan et al. (2019) and Hopenhayn et al. (2022) show that slowdowns in labor supply growth are a primary driver of declining startup rates, with entry suppressed while incumbent dynamics remain largely intact, which is precisely the pattern we document in the context of immigration enforcement. Our findings show that a specific federal policy can suppress the entry margin of business dynamism, connecting immigration enforcement to a macro debate about the long-run sources of declining firm formation and economic dynamism (Decker et al., 2020).

The remainder of the paper is organized as follows. Section 2 provides institutional background, Section 3 describes the data, and Section 4 outlines the empirical strategy. Section 5 reports the results, beginning with the local labor-supply first stage and then examining the number of firms and establishments, entry and exit, heterogeneity, hiring, entrepreneurship, and robustness. Section 6 concludes.

2 Institutional Background

2.1 Secure Communities: Design and Operational Mechanism

The 1996 Illegal Immigration Reform and Immigrant Responsibility Act substantially expanded the scope of interior immigration enforcement in the United States, increasing federal oversight and strengthening cooperation between federal and local law enforcement agencies.¹ This expansion raised the expected probability of detection and removal, altering local labor supply conditions.

Secure Communities, introduced in 2008, became the central interior enforcement program of this period. Its design differs from earlier programs by relying on automated information sharing between federal agencies. Under SC, when an individual is booked into a participating county jail, their fingerprints are transmitted to the Federal Bureau of Investigation and automatically checked against the Department of Homeland Security’s immigration databases. If the individual is identified as potentially removable, ICE is notified.²

¹It expanded civil penalties for unlawful presence, widened the use of expedited removal procedures, and, most relevant for this study, introduced Section 287(g), which authorizes formal agreements allowing state and local agencies to perform specified immigration enforcement functions under federal supervision. State-level omnibus immigration laws further extended enforcement through coordinated legislative efforts, including mandates on employment verification, restrictions on access to public benefits, limits on access to driver’s licenses and certain educational benefits, and expanded local law enforcement authority.

²Following a match, ICE may issue an immigration detainer requesting that the local jail hold the individual for up to 48 additional hours beyond their scheduled release to allow federal authorities to assume custody.

Because this data-sharing is triggered automatically once the required infrastructure is in place, activation does not require affirmative policy adoption by local jurisdictions.

By increasing the probability that an arrest leads to detention and removal, SC raises the expected cost of residing and working without legal status through several behavioral channels. First, actual deportations directly remove workers from the local labor force (Cox and Miles, 2013; East et al., 2023; Alsan and Yang, 2024). Second, the elevated risk of detection induces voluntary out-migration among undocumented residents who relocate to jurisdictions with lower enforcement intensity (Bohn et al., 2014; Amuedo-Dorantes et al., 2019; Smith, 2026). Third, SC may deter new inflows of undocumented immigrants who would otherwise have entered the local labor market (Amuedo-Dorantes et al., 2019; Hanson and Spilimbergo, 1999). In particular, the design of SC implies stronger effects along margins that depend on new hiring and business formation, and in sectors with greater reliance on immigrant labor.

2.2 Federal Rollout and the Source of Identifying Variation

Secure Communities was implemented through a phased national rollout between 2008 and 2013, achieving near-universal county coverage by the end of that period. Activation occurred at the county level as data-sharing capabilities between local law enforcement agencies and federal databases were established and as federal enforcement capacity expanded. In contrast to earlier programs such as 287(g), which relied on voluntary agreements initiated by local jurisdictions, requiring counties to actively apply, negotiate terms, and enter into formal memoranda of agreement with ICE, SC was deployed through a centralized federal process and did not require counties to apply for participation or take any affirmative steps toward adoption. This distinction is important for identification: because 287(g) adoption required local initiative, its timing plausibly reflected local political preferences, enforcement priorities, and immigrant population characteristics, making it a poor source of quasi-experimental variation. SC activation, by contrast, was determined by federal infrastructure deployment decisions, making its timing substantially more plausibly exogenous to local economic conditions (Cox and Miles, 2013; East et al., 2023).

Although the rollout was not randomized, its timing was driven primarily by federal operational considerations, including technical readiness and administrative capacity, rather than contemporaneous local economic conditions. Conditional on county and time fixed effects, we derive identification from differences in the timing of federal activation across counties. A remaining concern is that earlier activation was concentrated in counties with larger immigrant populations, border or metropolitan enforcement priorities, and potentially

different exposure to the Great Recession. These features do not mechanically invalidate the design; they are partly a consequence of where the federal system first had operational and enforcement capacity, but they mean that SC timing should not be interpreted as randomly assigned.

We address this concern in three ways. First, in our regression specification, we control for a rich set of time-varying county and state characteristics, including other concurrent immigration enforcement policies, local labor demand conditions, and housing market exposure, which absorbs observable confounders that might correlate with both activation timing and business outcomes. Second, we report event-study estimates and formal pre-period tests, distinguishing outcomes with clean pre-trend evidence from those where the diagnostics are more mixed. Third, and most directly, we test empirically whether pretreatment trends in local business dynamics, demographic changes, unemployment, housing conditions, and political characteristics predict the timing of SC activation, and find no systematic relationship, as we document in Section 4. While targeting on unobservables cannot be fully ruled out, the combination of centralized rollout, high-dimensional fixed effects, and flat outcome-specific pre-trend provides substantial support for the identifying assumption.

Figure 1 illustrates the geographic expansion of SC across U.S. counties during the rollout period, with activation proceeding unevenly across counties before reaching roughly 94% of counties by 2012. SC was discontinued in 2014 and replaced by the Priority Enforcement Program, which narrowed enforcement to individuals with serious criminal convictions, marking the end of SC as a nationally uniform policy.

3 Data

We use data from 2005 to 2012, which includes several pre-treatment years prior to the initial rollout of SC in 2008. We restrict the estimation sample to this window for two reasons. First, the interaction-weighted estimator of [Sun and Abraham \(2021\)](#), detailed in Section 4, identifies each cohort-specific treatment effect using only not-yet-treated or never-treated counties as the comparison group at each calendar date.³ By 2012 this pool is nearly exhausted: only 176 counties (5.6% of the sample) remain untreated, and extending the window further reduces this substantially (Appendix Table A.1), leaving too few untreated counties to support reliable inference at longer event-time horizons. Second, as discussed in Section 2, SC was discontinued in 2014 and replaced by the Priority Enforcement Program.

³Without a not-yet-treated or never-treated comparison group, estimation would require reintroducing already-treated units as controls, the source of bias the estimator is designed to avoid ([Goodman-Bacon, 2021](#)).

Because this redefines the treatment itself rather than merely eroding the comparison group, extending the sample past 2012 would mix a stable policy regime with a fundamentally different one. Restricting the sample to 2005-2012 for estimation therefore satisfies both requirements: a comparison group sufficient to identify cohort-specific effects, and a treatment variable with constant policy meaning throughout the estimation window. For all datasets, we restrict the sample to the contiguous United States. Below, we describe our main data sources.

3.1 Data Sources

3.1.1 Business Dynamics Statistics

Our primary dataset for outcome variables is the Business Dynamics Statistics (BDS), published annually by the U.S. Census Bureau and constructed from the Longitudinal Business Database (LBD), which links administrative records on employer establishments over time. The BDS reports non-farm establishment and employment counts, along with entry and exit flows and job creation and destruction, at the county-sector-year level. Sectors correspond to 2-digit NAICS industry classifications. The BDS sample contains 479,712 county-sector-year observations for our sample period.

Our main BDS outcomes are the number of firms, the number of establishments, establishment entry, and establishment exit.⁴ We examine job creation and job destruction separately in the hiring subsection. In the BDS, these measures are net employment changes at the establishment level, not gross worker flows. If an establishment grows in employment over the year, it contributes to job creation by the amount of that increase. If it shrinks, it contributes to job destruction by the amount of that decrease. Worker turnover within a stable headcount, such as one worker leaving and another being hired, does not appear in these measures.⁵ The data also provide rich heterogeneity by firm size and firm age, which we exploit to examine differential responses across these dimensions.

The BDS suppresses cells with establishment or employment counts below a disclosure threshold, resulting in missing values for some county-sector-year cells. We address the potential for selection bias from this suppression in Section 5.7.2.

⁴We follow the BDS convention throughout. An *establishment* is a single physical location where business is conducted. A *firm* is a legal entity that may own one or more establishments.

⁵To further decompose local job dynamics, we distinguish between flows arising from establishment births and deaths and those from continuing establishments. Specifically, job creation is decomposed into job creation from establishment births and job creation from expansions among continuing establishments, while job destruction is decomposed into job destruction from establishment deaths and job destruction from contractions among continuing establishments.

3.1.2 American Community Survey

We use the American Community Survey (ACS) to track population composition and incorporated self-employment. For the 2005-2012 entrepreneurship panel, we use official ACS 1-year Public Use Microdata Sample files. We define incorporated self-employment using the class-of-worker category for self-employed workers whose businesses are incorporated, aggregate the weighted rates and counts at the Public Use Microdata Area (PUMA) level, and allocate PUMAs to counties using population crosswalks. For baseline measures of immigrant exposure used in the heterogeneity analysis, we use ACS 5-year data accessed via the Census API ([Walker and Herman, 2021](#)). County-level shares of immigrants with a high school education or less come from ACS summary tables on nativity by educational attainment (2006-2010), available directly at the county level without a crosswalk. Sector-level shares are computed from the 2009-2013 5-year PUMS, since industry detail is not available in the summary tables.

3.1.3 Additional Datasets

To characterize differences in labor market frictions across sectors, we use the Quarterly Workforce Indicators (QWI), a Census Bureau dataset derived from the Longitudinal Employer-Household Dynamics (LEHD) program, which links non-farm employer and employee administrative records and provides measures of employment, earnings, hires, separations, and turnover. We use these data to construct baseline sectoral turnover rates at the county-sector level using the 2005-2007 data, which serve as a proxy for hiring frictions and allow us to examine whether enforcement effects are stronger in high-turnover sectors.

We use the 2005-2012 Census Nonemployer Statistics to measure the annual number of businesses with no paid employees in each county. The all-industry county totals provide an administrative check on the broader own-account business margin and complement the incorporated self-employment measures from the ACS.

We also construct a measure of state-level housing boom exposure using data from the Federal Housing Finance Agency (FHFA). Specifically, following [East et al. \(2023\)](#), we compute the population-weighted average county-level FHFA house price index within each state in 2000 and 2006, and define the percentage change as a time-invariant measure of housing market exposure. This measure is interacted with linear and quadratic time trends to flexibly capture differential housing cycle trajectories across states.

To control for other immigration enforcement policies, we compile data from several sources. Specifically, we collect information on the timing of 287(g) program adoption at the county and state levels from [Kostandini et al. \(2014\)](#), E-Verify implementation from

Orrenius and Zavodny (2015), state-level omnibus immigration laws from Allen and McNeely (2017) and Luo and Kostandini (2023), and sanctuary policy adoption from Martínez-Schuldt and Martínez (2019), where for each county we identify the earliest adoption date of any sanctuary policy at the city, county, or state level. We verify these data against official records from ICE and the DHS.

Finally, we obtain state minimum wage data from the Benzippere Historical Minimum Wage Database (Vaghul and Zipperer, 2016), which we include as a control variable in our regressions.

3.2 Descriptive Statistics

Table A.2 reports pre-treatment descriptive statistics for both unweighted and population-weighted averages. Weighted averages are substantially larger throughout, as economic activity and immigrant populations are disproportionately concentrated in more populous counties. In what follows we discuss the weighted figures, which correspond to the estimator used in our regressions and are weighted by county population in 2000.

Weighted by baseline county population, the average county contains 20,837 firms and 24,507 establishments, with 3,125 establishment entries and 2,653 exits per year over 2005-2007, alongside 71,708 jobs created and 66,087 jobs destroyed. These magnitudes are consistent with the well-documented high pace of firm and job reallocation in the U.S. economy (Davis et al., 1998; Haltiwanger et al., 2013), where entry and hiring flows constitute a central margin of adjustment.

At the county-sector level, the unit of analysis in our regressions, the population-weighted average cell contains 1,122 firms and 1,320 establishments. This reflects a non-trivial density of economic activity even within narrowly defined local markets, comparable to the level of disaggregation used in studies of local labor demand shocks and business dynamics (Autor et al., 2013; Greenstone et al., 2010). The richness of this variation allows us to exploit differential exposure across sectors within the same local labor market.

The population-weighted average non-citizen share is 8.1%, with a foreign-born low-education share of 8.6% and a Hispanic non-citizen share of 5.2%. This concentration is a key feature of the empirical setting, as it generates meaningful differences in exposure to immigration enforcement across counties, which we exploit in heterogeneity analyses. These features point to an environment with active firm and labor market adjustment and substantial heterogeneity in exposure to immigrant labor, which we leverage in the empirical strategy to identify the effects of immigration enforcement on local business dynamics.

4 Empirical Strategy

4.1 Empirical Specification

We estimate the effects of SC implementation on local business dynamics using a difference-in-differences framework with staggered treatment timing. We begin by outlining a standard difference-in-differences specification as a benchmark. A conventional three-way fixed effects model (TWFE) would take the following form:

$$Y_{c,j,t} = \beta SC_{c,t} + X'_{c,t}\phi + \mu_c + \lambda_t + \delta_j + \varepsilon_{c,j,t} \quad (1)$$

where $Y_{c,j,t}$ denotes the outcome variable for county c , sector j , and year t . The variable $SC_{c,t}$ is a binary indicator equal to one when SC is active for a majority of months in year t in county c .

The specification includes county fixed effects (μ_c), year fixed effects (λ_t), and sector fixed effects (δ_j). County fixed effects absorb all time-invariant county characteristics. Year fixed effects absorb aggregate time shocks common to all counties. Sector fixed effects absorb time-invariant differences across industries, such as baseline differences in entry, exit, and employment dynamics.⁶

The vector of county- and state-level controls, $X_{c,t}$, includes indicators for county-level 287(g) agreements, state-level 287(g) agreements, E-Verify mandates, Omnibus Immigration Bills, sanctuary policies,^{7 8} the state minimum wage, a Bartik labor demand index that captures sector-specific local demand shocks (Bartik, 1991; Autor et al., 2013),⁹ and a state-level housing boom exposure measure interacted with linear and quadratic time trends.¹⁰ Because our sample period spans the Great Recession, these last two variables flexibly capture differential exposure to both housing market shocks and broader macroeconomic demand fluctuations across states. As an additional check, we show in Section 5.7 that our results are robust to replacing year fixed effects with commuting-zone-by-year, region-by-year, and Cen-

⁶Following East et al. (2023), we do not include state-by-year fixed effects, because SC was activated across all counties in some states simultaneously, so that state-by-year fixed effects would absorb much of the identifying variation in the SC rollout.

⁷Sanctuary policies can be implemented at the city, county, or state level. For each county, we identify the earliest adoption date of any sanctuary policy at any of these levels and define annual treatment as an indicator equal to one if the policy is in effect for at least six months of a given year.

⁸See Appendix B for description of these immigration policies.

⁹See Appendix C for details on its construction.

¹⁰We construct this variable as follows: we construct the population-weighted average county-level FHFA house price index within each state, computed for 2000 and 2006, where the percentage change defines a time-invariant state-level housing boom exposure. This shock is interacted with linear and quadratic time trends to flexibly capture differential housing cycle trajectories across states.

sus division-by-year fixed effects, as well as to including state-by-year fixed effects directly, which further absorb any differential regional recovery from the Great Recession.

The error term $\varepsilon_{c,j,t}$ is clustered at the county level to account for serial correlation and arbitrary within-county dependence over time. All regressions are weighted by county population in 2000 to account for differences in county size and to ensure that estimates reflect effects on the average resident rather than the average county.¹¹

Equation (1) provides a familiar starting point for the empirical design. However, in settings with staggered treatment timing and heterogeneous treatment effects, the two-way fixed effects estimator can produce biased estimates (De Chaisemartin and d’Haultfoeuille, 2020; Callaway and Sant’Anna, 2021; Goodman-Bacon, 2021; Sun and Abraham, 2021; Borusyak et al., 2024). In staggered adoption settings, TWFE implicitly uses already-treated counties as controls for newly treated ones, and when treatment effects vary across cohorts or over time, this generates negative weights on some comparisons, potentially biasing the aggregate estimate (Goodman-Bacon, 2021). We therefore rely on the interaction-weighted estimator of Sun and Abraham (2021), which restricts all comparisons to not-yet-treated counties and aggregates cohort-specific effects using non-negative weights, yielding an interpretable average treatment effect that is robust to arbitrary treatment effect heterogeneity. In the robustness section, we show that results are stable using the imputation estimator of Borusyak et al. (2024).

4.2 Average Treatment Effect and Event Study Estimates

To estimate dynamic treatment effects and assess pre-trends, we implement an event study using the interaction-weighted estimator of Sun and Abraham (2021), which allows treatment effects to vary across cohorts and over time. The specification is:

$$Y_{c,j,t} = \sum_{g \in \mathcal{G}} \sum_{k \neq -1} \beta_{g,k} \mathbf{1}(E_c = g) \mathbf{1}(t - g = k) + X'_{c,t} \phi + \mu_c + \lambda_t + \delta_j + \varepsilon_{c,j,t}, \quad (2)$$

where E_c denotes the first year in which SC is active for a majority of months in county c . The term $\mathbf{1}(E_c = g)$ identifies counties first treated in year g , and $\mathbf{1}(t - g = k)$ indicates event time k relative to treatment. The coefficients $\beta_{g,k}$ capture cohort-specific treatment effects at event time k , with $k = -1$ omitted as the reference period.¹²

¹¹We also report unweighted estimates in the robustness section. These assign equal weight to each county and allow us to assess whether the results are driven disproportionately by more populous counties.

¹²Counties first treated in 2013 or later are classified as never-treated, following the standard practice in staggered difference-in-differences designs of using not-yet-treated as the comparison group (Sun and Abraham, 2021; Callaway and Sant’Anna, 2021). Table A.1 reports the distribution of counties across activation cohorts.

For presentation, we report the interaction-weighted averages of these cohort-specific effects at each event time,

$$\beta_k = \mathbb{E}_g[\beta_{g,k}],$$

which summarize the average treatment effect at relative time k .

The event window spans $[-4, +3]$ relative to first activation.¹³ Pre-treatment coefficients for $k \in \{-4, -3, -2\}$ provide a diagnostic for differential trends. Estimates close to zero with narrow confidence intervals would be reassuring, whereas failure to reject zero with wide intervals cannot rule out economically meaningful differences. Post-treatment coefficients trace the dynamic response of business outcomes following SC activation.

To summarize the overall effect of SC, we report an average treatment effect on the treated obtained by aggregating cohort- and event-time-specific treatment effects estimated using [Sun and Abraham \(2021\)](#). Formally,

$$ATT = \mathbb{E}_g [\mathbb{E}_{k \geq 0} [\beta_{g,k}]], \quad (3)$$

where $\beta_{g,k}$ denotes the treatment effect for cohort g at event time k . In practice, this corresponds to a weighted average of post-treatment event-time coefficients, where weights reflect the relative frequency of treated observations across cohorts and event times. Specifically, cohorts with more counties and more post-treatment periods observed within the sample window receive greater weight in the aggregation.

4.3 Identification

Our empirical strategy compares within-county changes in business outcomes around the timing of SC activation to those in counties that have not yet been treated, while controlling for county, year, and sector fixed effects and a rich set of policy and economic controls. We interpret the estimates as effects for treated county-sector cells exposed during the federal rollout, rather than as a universal elasticity of business dynamics with respect to all forms of immigration enforcement. Identification exploits staggered variation in the timing of SC activation across counties.

Formally, the identifying assumption is a generalized parallel trends condition. Let $Y_{c,j,t}(0)$ denote the untreated potential outcome. Then, for all event times $k < 0$,

$$E[Y_{c,j,t}(0) - Y_{c,j,t-1}(0) \mid E_c = g, X_{c,t}] = E[Y_{c,j,t}(0) - Y_{c,j,t-1}(0) \mid E_c = g', X_{c,t}], \quad (4)$$

¹³The asymmetric window reflects the structure of the data, with four pre-treatment periods available from the 2005 sample start and post-treatment periods ranging from one to three depending on the activation cohort.

for any cohorts g and g' . This assumption implies that, absent SC, early- and late-adopting counties would have followed similar trends in business dynamics. Federal infrastructure deployment rather than local economic conditions determined activation timing, which reduces the scope for local economic trajectories to set treatment dates. Table A.3 also shows little systematic relationship between pre-treatment changes in observed county characteristics and activation timing.¹⁴

5 Results

5.1 Enforcement and Local Labor Supply

Since Secure Communities targets individuals rather than businesses directly, this section verifies the policy’s first-stage effect on the local immigrant workforce before tracing its impacts on firms. Using the Sun and Abraham (2021) event study design applied to ACS county-level population shares, restricted to counties with population above 65,000, we examine two measures: the non-citizen share and the Hispanic non-citizen share, the latter of which captures the group most exposed to SC deportations (Passel et al., 2014). Figure 2 reports the event study estimates for these two measures. Pre-treatment coefficients are flat for both measures, supporting the parallel trends assumption. Following activation, both measures decline and grow more negative through the third post-treatment year, with the sharpest drop observed for the Hispanic non-citizen share.

Table A.4 reports the aggregate ATTs, which confirm this pattern. The non-citizen share falls by 0.218 percentage points, a 2.5% reduction. The Hispanic non-citizen share falls by 0.211 percentage points, a 3.6% reduction significant at the 5% level. Because the non-citizen category mixes low-risk legal residents with high-risk undocumented immigrants, the true effect on the undocumented population is likely larger than these aggregate figures suggest. If roughly half of non-citizens nationally were undocumented circa 2010, the estimated decline could translate into approximately a 5.0% drop in the undocumented population, bracketing the shock between 2.5%, if spread evenly across non-citizens, and 5.0%, if concentrated among the undocumented. This range sits just around East et al. (2023), who find that SC reduced undocumented employment by roughly 6%.

Two caveats shape how the remaining results should be interpreted. First, these are reduced-form policy effects rather than structural labor supply elasticities, since enforcement also reduces local consumer demand and raises uncertainty among those who remain.

¹⁴The only statistically significant predictor in Appendix Table A.3 is establishment growth, which has a very small coefficient and predicts earlier activation in faster-growing counties.

Second, the measured population decline likely understates the true labor supply shock, because enforcement further discourages labor force participation and job mobility even among immigrants who stay. This roughly 2.5 to 5% contraction in the immigrant population establishes the magnitude of the shock that the paper traces through subsequent effects on business formation and dynamics.

5.2 The Effects on the Number of Establishments and Firms

Having established the size of the labor supply shock, we turn to its consequences for local businesses. We first examine the effects of SC on the number of establishments and firms operating within a county. These stock measures incorporate entry, exit, expansion, and contraction, so a decline indicates a net reduction in local business activity rather than a reallocation across gross margins.

Table 1 reports aggregate ATTs from the interaction-weighted estimator of [Sun and Abraham \(2021\)](#), estimated under the baseline specification of Section 4. We estimate regressions in levels and express coefficients as a percentage of the pre-treatment mean of the outcome in treated counties.¹⁵ Because the [Sun and Abraham \(2021\)](#) aggregate ATT is a weighted average of cohort-by-period effects, we construct that mean from each cohort’s own pre-treatment level and aggregate it using the same cohort weights the estimator applies, so that numerator and denominator describe the same population.¹⁶

The results show that SC activation reduces the number of establishments by 2.9% relative to that mean. The number of firms falls by 3.4%. Both estimates are significant at the 1% level. Figure 3 presents the corresponding event studies. The pre-treatment coefficients are all economically small, similar in magnitude, and statistically indistinguishable from zero, consistent with parallel trends prior to activation. Following activation, the estimates for both firms and establishments turn negative at $k = 0$ and grow larger in magnitude with time, though the longer-horizon coefficients are not statistically significant at the 95% level.¹⁷

¹⁵Log-like transformations such as $\log(1 + y)$ or $\text{arcsinh}(y)$ yield scale-dependent estimates in settings where treatment affects the extensive margin, as their interpretation depends on the units of the outcome ([Chen and Roth, 2024](#)). We therefore report effects in levels throughout.

¹⁶Appendix P reports how the percentages move under alternative constructions of the baseline.

¹⁷We interpret this growth with caution. As discussed in Section 3, our sample ends in 2012 because the comparison group of not-yet-treated counties is nearly exhausted by then. A consequence is that later event-time coefficients are identified by progressively fewer, earlier-activating cohorts that are geographically distinctive (e.g., $k = +3$ is identified by only the 51 counties of the 2009 cohort, 94% of which are border or southern states). We therefore rest our interpretation on $k = 0$ and $k = +1$ and treat later horizons as suggestive evidence from early adopters rather than the average treated county’s longer-run trajectory. Appendix D explains this in full.

The magnitude of the decline is best interpreted relative to the underlying shock. Relative to the rescaled population decline of 5.0% documented in Subsection 5.1, the observed declines of 3.4% for firms and 2.9% for establishments imply elasticities of approximately 0.68 and 0.58, respectively. Under free entry, the number of firms scales approximately one-for-one with effective market size, and the size of the local labor force is a first-order determinant of firm entry (Hopenhayn et al., 2022; Karahan et al., 2019). The elasticities we estimate sit at or below this benchmark of unity, indicating that the local business population contracts by somewhat less than the labor pool that sustains it.

The proportional decline is somewhat larger for firms than for establishments. A county’s firm count rises only when a firm new to the county appears, whether newly created or an existing multi-county firm opening its first local establishment. The establishment count, by contrast, can also rise when an incumbent firm opens an additional location, independent of any new firm entering the county. If SC operates primarily by suppressing entry rather than incumbent activity, firms should absorb the full force of that suppression, while establishments are partially cushioned by continued expansion among incumbents that remain in operation. The larger proportional decline in firms relative to establishments is therefore consistent with the entry margin playing a central role in the aggregate decline, a possibility we directly test next.

5.3 Establishment Entry and Exit

The decline in the number of establishments documented in Subsection 5.2 is a net change, which could reflect fewer establishments opening, more establishments closing, or both, and a wave of closures and a shortfall of openings involve different businesses and warrant different policy responses. This subsection separates the two margins. Table 2 reports aggregate ATTs from the interaction-weighted estimator of Sun and Abraham (2021) for establishment entry and establishment exit.

The two margins respond asymmetrically. SC activation reduces establishment entry by 9.5% relative to the pre-treatment mean, significant at the 5% level. The exit coefficient is statistically indistinguishable from zero, with a point estimate that, if anything, points toward fewer exits (1.3% of the pre-treatment mean). Enforcement operates by preventing establishments from opening rather than by closing those already in place. This asymmetry is precisely what a model of free entry with recruiting frictions predicts: an entrant must assemble an entire workforce from a labor pool that enforcement has thinned, whereas incumbents remain largely insulated by the matches they already hold. Appendix E formalizes

this logic, and Section 5.5 traces it through the hiring margin; the same force that deters entry gives incumbents no comparable incentive to exit.

Figure 4 presents the corresponding event studies. Pre-treatment coefficients for both entry and exit are small in magnitude, show no discernible trend, and are statistically indistinguishable from zero, consistent with parallel trends prior to activation. Following activation, entry turns negative and grows larger in magnitude through the post-treatment window, while exit shows no comparable pattern, remaining close to zero through the first two post-treatment years before drifting modestly negative at longer horizons. Estimates at longer horizons become less precise as fewer counties contribute to those event-time cells.

The flow margins account for the decline in the number of establishments. Because the annual change in the number of establishments equals entries minus exits, an entry shortfall of roughly 18 establishments per cell per year, with exit unchanged, implies approximately 18 fewer establishments in the activation year and 36 fewer the following year. Averaging across the four post-treatment years yields a shortfall of approximately 45, closely matching the aggregate ATT of 45.7 reported in Subsection 5.2. This calculation constitutes an order-of-magnitude accounting check rather than a decimal-level validation, as it abstracts from differences in the number of observed post-treatment years across cohorts and sets the statistically insignificant exit response to zero. The decline in the number of establishments and firms therefore reflects a shortfall of arrivals, and this shortfall accumulates for as long as entry remains depressed.

How does the entry decline compare to the first-stage population effect? Establishment entry falls by 9.5% while the non-citizen share of the population falls by 2.5%, an implied ratio of roughly four. The comparison overstates the relevant response for two reasons. First, scaled to the population actually at risk of removal, the first stage is closer to 5.0% (Subsection 5.1), which alone halves the ratio. Second, entry is an annual flow, whereas the population share is a level; transition flows can respond more sharply while the resulting change in the number of active businesses accumulates over time. Relative to the rescaled population decline, entry falls by roughly 1.9 times as much in percentage terms. The like-for-like elasticities in Subsection 5.2, at 0.58 to 0.68, sit below the free-entry benchmark of one. We verify that this comparison is not driven by comparing outcomes measured at different levels of aggregation. Collapsing the entry regression from the county-sector-year panel to a county-year panel, matching the level at which the population first stage is estimated, yields a decline of 8.4%, nearly identical to the sector-level estimate of 9.5% (see Appendix F). The entry decline therefore holds at the county level and is not inflated by the additional sector dimension.

The entry shortfall is not distributed uniformly across businesses. Entry declines by 10.3% among small establishments and by 11.4% among young establishments. Entry is statistically indistinguishable from zero for medium, large, mature, and old categories (Appendix G). These within-category patterns are descriptive, as we do not estimate formal interaction terms comparing subgroup effects.

The adjustment falls on entry because entrants are the businesses most exposed to the labor supply shock documented in Subsection 5.1: an entrant must assemble an entire workforce from a labor pool that enforcement has thinned. The subsections that follow examine where the four business outcomes concentrate (Subsection 5.4) and how the hiring margin responds (Subsection 5.5).

5.4 Exposure and the Concentration of Effects

The aggregate results establish the average response. We next examine whether it is concentrated in places and sectors with greater exposure to immigrant labor. Every heterogeneity table reports the same four outcomes: the number of firms, the number of establishments, establishment entry, and establishment exit. We divide each exposure measure at its median and estimate Sun and Abraham (2021) ATTs separately by subgroup. Comparing subgroup point estimates or their significance separately would not establish that the two differ, so we test each difference directly using a county-cluster bootstrap and report the results in Table A.5.¹⁸

Sector immigrant intensity. We begin with the most direct measure of exposure: how reliant a sector is on immigrant workers to begin with. Table 3, Panels A and B, splits sectors by baseline immigrant employment share. Table A.6 reports the immigrant share by sector underlying this split. In high-immigrant-share sectors, SC activation reduces the number of firms by 4.5% and the number of establishments by 3.8%. Establishment entry falls by 13.6%. All three estimates are significant at the 5% level, while establishment exit is statistically indistinguishable from zero. In low-immigrant-share sectors, the four estimates are smaller in magnitude and statistically indistinguishable from zero. The difference between the two groups is statistically significant for the number of firms and the number of establishments (both at the 5% level) and for entry (10%), so the concentration in immigrant-intensive sectors reflects a genuine difference in treatment effects rather than a difference in precision.

¹⁸The Sun and Abraham (2021) estimator aggregates cohort-by-time ATTs and does not admit an interaction term, and the two sector subsamples are drawn from the same counties, so combining their standard errors under an independence assumption is invalid. We therefore bootstrap the difference by resampling counties with replacement and re-estimating both subgroup ATTs on each resample, so that the bootstrap distribution of the difference carries the covariance between them.

County immigrant share. A sector’s reliance on immigrant labor is a within-county measure; it says nothing about how exposed the surrounding county is. Panels C and D split counties by their baseline immigrant population share instead, asking whether the same pattern holds at the level of place rather than industry. In high-share counties, the number of firms falls by 3.4%, the number of establishments falls by 2.9%, and entry falls by 9.2%. Exit is statistically indistinguishable from zero. In low-share counties, the estimates for the number of firms and establishments are close to zero. Entry and exit are both statistically significant but measured against much smaller pre-treatment means. Thus, low-share counties do not reproduce the negative count response, but their entry and exit estimates are not uniformly null. Sector and county exposure therefore point in the same direction despite capturing different dimensions of exposure, suggesting that the concentration is not driven solely by one particular way of defining exposure.

Sector turnover. Immigrant share alone does not explain why a sector’s exposure translates into a business response; it also matters how much labor market friction that sector faces in recruiting. Sectors with high worker turnover must recruit continuously to replace separations and sustain production, so a labor supply shock should bind more tightly where recruiting is a recurring necessity rather than an occasional one. Table A.7 reports the underlying rates, and Figure A.1 shows that sector immigrant shares and baseline turnover are strongly correlated (Pearson $r = 0.77$). We therefore interpret turnover as a characteristic of the same exposed sectors rather than as an independent mechanism.

Table 4 shows a descriptive gradient across turnover groups. In high-turnover sectors, SC reduces the number of firms by 4.0%, the number of establishments by 3.4%, and entry by 11.3%. Exit is statistically indistinguishable from zero. In low-turnover sectors, the corresponding estimates are 1.8%, 1.5%, and 6.8%; each is marginally significant at the 10% level. These results are consistent with larger adjustment in sectors that depend on continuous recruitment. The high-minus-low turnover difference is significant at the 10% level for the number of firms and the number of establishments, though not for entry.

Tradability and local demand. *Tradability and local demand.* A negative labor supply shock is not the only channel through which enforcement could suppress firm entry. If immigrant households leave a county, local consumer spending may fall as well, and businesses that depend on local demand would contract for that reason independent of any change in the workers available to hire. Tradability lets us assess this channel: non-tradable sectors, such as restaurants, retail, and personal services, sell to local consumers and are exposed to

a decline in local demand, whereas tradable sectors sell to broader markets and are largely insulated from a local population decline on the demand side.¹⁹

Table 5 reports the pooled benchmark in Panel A and re-estimates the same specification by tradability in Panels B and C. In non-tradable sectors the number of firms falls by 4.0% and the number of establishments by 3.3%, both significant at the 1% level and both larger than the corresponding pooled declines of 3.4% and 2.9%. In tradable sectors the same estimates are 0.3% and 0.4% and are statistically indistinguishable from zero. This pattern is consistent with a local-demand channel operating alongside labor supply: non-tradable sectors, which depend directly on local consumer spending, show the largest declines, while tradable sectors, insulated from local demand, show essentially none.

The non-tradable-minus-tradable difference is not statistically significant for the number of firms ($p = 0.14$), the number of establishments ($p = 0.17$), or entry ($p = 0.23$), as reported in Table A.5, so we do not treat this comparison as formally establishing that a demand channel is separable from labor supply. The pattern is nonetheless suggestive: non-tradable estimates exceed the pooled benchmark while tradable estimates sit near zero, consistent with local consumption spending amplifying the effects of enforcement where businesses depend most on it.

Together, the heterogeneity results locate the average reductions in immigrant-intensive and high-turnover settings, and the formal tests support that reading: the immigrant-intensity difference is significant at the 5% level for the number of firms and establishments, and the turnover difference at the 10% level. The tradability difference is not significant, so the local-demand channel remains suggestive rather than established. None of these splits isolates a single mechanism, since immigrant labor supply, local consumption, and network exposure all vary along the same dimensions. What the tests establish more clearly is that the average effects are concentrated along exposure margins most closely associated with immigrant labor dependence, while the evidence for an additional local-demand channel is suggestive rather than conclusive.

5.5 Does the Hiring Margin Respond?

Business counts capture one margin of adjustment, but establishments already operating can also respond by hiring less. If enforcement operates by thinning the pool of workers firms recruit from, hiring should contract while separations show no systematic response, mirroring the entry-exit asymmetry documented in Subsection 5.3. The BDS lets us observe these margins directly. While East et al. (2023) show that SC reduced employment among

¹⁹Tradable sectors are agriculture, mining, and manufacturing, following Burstein et al. (2020); all remaining sectors are non-tradable.

both likely undocumented immigrants and U.S.-born workers, worker-level data alone cannot distinguish whether firms adjusted through lower hiring, higher separations, reduced entry, or increased exit.

Table 6 reports Sun and Abraham (2021) aggregate ATTs. Job creation falls by 10.3% of the pre-treatment mean, significant at the 5% level. Job destruction is statistically indistinguishable from zero, a 2.2% change relative to the pre-treatment mean: hiring contracts, and separations do not rise.

The model in Appendix E predicts this asymmetry should hold unevenly within hiring itself: entrants face the full recruiting friction, assembling an entire workforce from scratch, while incumbents have already matched with workers and face recruiting costs only on replacement hires and marginal expansion. Table 7 tests this by decomposing job flows into establishment births, deaths, and continuing establishments, a distinction unavailable in worker-level data. The decline in job creation is driven almost entirely by continuing establishments: job creation at continuers falls 14.3%, while job creation at births declines modestly and is not statistically distinguishable from zero. Job destruction at deaths changes by only 1.1% and is insignificant, with similarly small and imprecise changes among continuers.

Of a combined decline of 415.8 jobs per county-sector-year cell, continuing establishments account for 354.5 and births for 61.3.²⁰ Roughly 85 percent of the job-creation response therefore occurs at establishments already operating, though the births component is imprecise enough that this share could plausibly range from about 70 percent to essentially the entire decline. This decomposition refines the entry result rather than contradicting it: the decline in the *number* of businesses operates through entry, while the decline in *jobs* operates mainly through incumbents expanding less.

The muted births estimate is not inconsistent with the 9.5% entry decline. If new establishments were of unchanged average size, the entry decline would imply a drop of roughly 145 jobs in births job creation, a figure that lies inside that estimate’s confidence interval. A direct regression of job creation per new establishment on activation is itself statistically indistinguishable from zero.

Table A.9 splits the continuer decomposition at the median sector turnover rate, as the model predicts the incumbent response should be strongest where replacement hiring exposes firms to the recruiting friction on an ongoing basis. Job creation at continuing establishments falls by 16.2% in high-turnover sectors and 12.2% in low-turnover sectors, both significant; since turnover and immigrant share are strongly correlated (Pearson $r = 0.77$, Figure A.1),

²⁰The separately estimated components sum to -415.8 , slightly larger in magnitude than the -394.4 obtained by estimating total job creation directly; the discrepancy reflects BDS suppression thresholds that differ by outcome, so the three regressions are estimated on different samples (298,867, 404,201, and 410,707 observations, respectively).

these are the same sectors most exposed to the labor supply contraction. The split also clarifies the muted aggregate birth response: job creation at births falls by 12.4% in high-turnover sectors, close to the entry decline, while rising in low-turnover sectors.

Together, aggregate hiring contracts, the decline concentrates among continuing establishments rather than births or deaths, and there is no evidence of increased job destruction or exit. The labor supply shock operates through firms expanding less, not through firms severing existing matches, a distinction worker-level employment data alone cannot recover.

5.6 Entrepreneurship

Enforcement may reduce business formation directly by removing or deterring immigrant entrepreneurs, who start businesses at higher rates than natives (Kerr and Kerr, 2020; Fairlie and Lofstrom, 2015) and whose emigration has been shown to reduce firm creation in origin communities (Anelli et al., 2023). We test this channel using incorporated self-employment from the 2005-2012 ACS Public Use Microdata Sample, which captures owners who hire employees and formally register a firm, distinct from unincorporated own-account work (Levine and Rubinstein, 2017) and closer to the employer establishments in our main results. We construct county-year rates and counts separately for non-citizens, U.S. citizens, and all employed workers, allocating PUMA aggregates to counties with population crosswalks. Table 8 reports Sun and Abraham (2021) aggregate ATTs under the county-year baseline specification.

SC activation increases the non-citizen incorporated self-employment rate by 8.4% and the count by 23.0%, if anything the wrong sign for a story in which enforcement directly reduces immigrant business formation; both estimates are statistically indistinguishable from zero. The citizen rate falls by 2.4% and the count by 0.1%, and for all workers the rate changes by -1.2% and the count by $+2.7\%$; these too are statistically indistinguishable from zero. The ACS therefore provides no evidence that declining incorporated self-employment explains the entry decline, and the non-citizen point estimates give no support to a direct-removal channel in particular.

We complement this with Census Nonemployer Statistics, an administrative count of businesses with no paid employees that avoids ACS sampling error, though it does not distinguish owners by citizenship. If enforcement displaced activity into informal self-employment rather than reducing entrepreneurship outright, nonemployer counts should rise instead. Column (7) of Table 8 shows SC activation changes nonemployer establishments by -1.9% of the pre-treatment mean, a statistically null estimate and, if anything, of the wrong sign for a

displacement story. Neither margin shows evidence of reduced formal entrepreneurship or displacement into informality.

These null results support the paper’s mechanism. The population targeted by SC is overwhelmingly undocumented and faces substantial barriers to operating formal employer businesses, so enforcement’s effect on entry more plausibly runs through the labor supply available to eligible entrepreneurs than through a reduction in immigrant business ownership itself. This distinguishes our channel from the one emphasized in the immigrant-entrepreneurship literature, where the immigrant is the business owner rather than the employee. We cannot rule out economically meaningful changes, since the confidence intervals are wide, especially for non-citizen self-employment, but the point estimates show no common decline across citizenship groups or business definitions, consistent with entry contracting because entrepreneurs face a thinner labor pool rather than because enforcement removed the entrepreneurs themselves.

5.7 Robustness Checks

5.7.1 Labor Market Aggregation and Spatial Spillovers

County-level estimates could reflect geographic displacement rather than a true labor-market effect if enforcement shifts workers and businesses to neighboring counties. Table A.10 addresses this with commuting-zone-by-year fixed effects (Panel B) and aggregation to the commuting-zone-sector-year level (Panel C), against the baseline (Panel A). Under commuting-zone-by-year fixed effects, firms fall 2.2%, establishments 1.8%, and entry 9.4%, with exit null. At the commuting-zone level, firms fall 5.3%, establishments 5.0%, and entry 13.9%, with exit again null. The negative count and entry estimates therefore survive both within-market comparisons and aggregation to the labor-market level.

Appendix Table L.1 adds population-weighted SC coverage of nearby counties, following Clarke (2017) and Butts (2024). With both cross-county exposure measures included, own-SC estimates remain negative for firms, establishments, entry, and exit; the counts remain significant while entry and exit become imprecise. Cross-CZ exposure is positively associated with firm and establishment counts in untreated counties, consistent with partial displacement. Because untreated, unexposed counties fall to just 245 by 2011 and zero by 2012, we read this as evidence that some count activity relocates across labor markets while the own-county count decline persists after correction; it is less informative about entry.

5.7.2 Missing Data and Selection Bias

The BDS data suppresses cells below a disclosure threshold: 12.9% of firm/establishment observations, 36.5% of entry, and 36.1% of exit are missing (Table A.11). If enforcement-induced declines push cells below this threshold, differential missingness could bias the estimates. We assess this with three tests.

First, we restrict to county-sector cells whose outcome is observed in every year (Table A.12). This is a demanding restriction for the flow outcomes, retaining 80% of cells for the counts but under 30% for entry and exit. The counts remain negative and significant at the 1% level and somewhat larger in magnitude than in the full sample, at 3.9% for firms and 3.2% for establishments; entry remains significant at 7.9%, and exit stays statistically indistinguishable from zero. Differential attrition therefore does not generate the results, and if anything the full-sample estimates are the more conservative of the two. Second, we use suppression itself as the outcome (Table A.13): SC activation’s effect on suppression probability is insignificant for firms, establishments, and entry, and significant only for exit. This works against rather than for our findings: if enforcement suppresses marginal exit cells, the true exit effect could exceed our (statistically null) estimate, reinforcing rather than undermining the entry-exit asymmetry. Third, trimming the smallest 1%, 5%, and 10% of cells by baseline establishment count (Table A.14) leaves the firm, establishment, and entry estimates negative and significant at every threshold, with exit null throughout. Together, these tests indicate that BDS suppression does not bias the main results.

5.7.3 Alternative Estimator and Fixed Effects

Table A.15 re-estimates the two outcomes with the imputation estimator of Borusyak et al. (2024). Static ATTs are -54.1 firms (4.1%) and -54.4 establishments (3.4%), close to our preferred estimates. Figure A.2 reports the corresponding event studies for these two outcomes.

Replacing year fixed effects and state-level controls with state-by-year fixed effects (Table A.16) leaves firms, establishments, and entry negative and significant (falling 35.6, 37.9, and 17.9, respectively), with exit again null. This specification identifies the SC coefficient only from within-state, within-year timing differences.

Because SC activated earlier in southern and border counties, we also check whether the design captures differential regional recovery from the Great Recession rather than enforcement. Replacing year fixed effects with region-by-year and Census division-by-year fixed effects leaves the estimates stable and, if anything, more precise (Appendix H).

5.7.4 Additional Checks

The Online Appendix reports further checks, summarized here.

Selection on unobservables. [Oster \(2019\)](#) bounds ([Appendix I](#)) are large and positive for firms and establishments, large and negative for entry, and smaller and negative for exit.²¹

Spatial inference. Conley standard errors at 100-300 km ([Appendix J](#)) leave entry significant at 5% and the counts significant at 10%, with exit null. A spatial-noise placebo exercise ([Appendix K](#)) confirms this: two-sided rank p -values are 0.046 (entry), 0.109 (firms), 0.102 (establishments), and 0.874 (exit), placing entry in the placebo distribution’s tail while the counts are unusual but not conventionally significant. The own-SC estimates are also stable to controlling for Secure Communities coverage in neighboring counties and commuting zones ([Appendix L](#)), indicating that spatial spillovers from nearby treated counties are not driving the results.

Clustering, weighting, and cohort composition. The entry-exit asymmetry holds under three further checks ([Appendices M, N, and O](#)). Two-way county-and-year clustering removes significance from the firm and establishment estimates, but entry stays significant at 5% (county/state clustering) or 10% (two-way); exit is null throughout. Unweighted estimates remain negative and significant for firms, establishments, and entry, with exit null, though weighting changes the estimand so levels are not directly comparable. Dropping each activation cohort in turn leaves the counts negative and significant and entry negative in every subsample (significance 10%-1%), with exit null in every case.

5.7.5 Sensitivity to the Percentage Normalization

All percentage effects divide a level ATT by a pre-treatment baseline, so the reported magnitude depends on how that baseline is constructed. Because the [Sun and Abraham \(2021\)](#) aggregate ATT is a weighted average of cohort-by-period effects, we aggregate the baseline with the same cohort weights, so numerator and denominator describe the same population; early-activating counties are large, with roughly six times the baseline entry of late adopters, so pooling all treated counties equally would understate the denominator and overstate every percentage.

[Table P.1](#) reports our main outcomes under four denominators: pooling treated counties without cohort weights (column 2), our cohort-weighted baseline (column 3), and two alternatives that replace the single pre-treatment year with a three-year pre-treatment average and a fixed 2005-2007 window (columns 4-5), addressing the fact that the cohort-weighted

²¹Because controls barely change the fixed-effects R^2 , these ratios are mechanically unstable and should be read only as directional diagnostics.

baseline draws on different calendar years across cohorts during a period when entry was falling steeply.

Stock outcomes are insensitive to this choice, since firm and establishment counts moved little over the sample period. Flow outcomes are more sensitive: establishment entry ranges from -7.8% to -9.5% and job creation from -8.2% to -10.3% , reflecting the recession-era decline in entry. We report the cohort-weighted single-year baseline (column 3) throughout, both the conventional choice and the most conservative of the three corrected columns. Level estimates, standard errors, and inference are identical across all four columns; only the normalization differs.

6 Conclusion

Interior immigration enforcement reduces the presence of local businesses. Exploiting the staggered, federally determined rollout of Secure Communities across U.S. counties and county-sector-year Census administrative data, we estimate that activation reduces the number of establishments by 2.9%, the number of firms by 3.4%, and establishment entry by 9.5%, while exit is statistically indistinguishable from zero in the preferred specification. We interpret this evidence together with the stability of the count and entry estimates under commuting-zone aggregation, state-by-year fixed effects, trimming, and leave-one-cohort-out checks, while treating the null exit response with caution.

The declines in business counts and entry concentrate in immigrant-intensive and high-turnover sectors, in high-immigrant counties, and among small and young businesses. We test these cross-group differences directly rather than inferring them from separate-sample significance, and both the immigrant-intensity and turnover differences are statistically significant, while the tradability contrast is not. Business stocks do fall by more in nontradable than in tradable sectors even though the tradable bin is the more immigrant-intensive of the two, a ranking that a labor supply contraction alone would not produce, but on the evidence available we treat the local-demand contribution as suggestive rather than established and do not rank the two channels. The direct entrepreneurship evidence is null: incorporated self-employment and the Census count of nonemployer establishments are unchanged within the precision of the estimates, shifting the weight toward channels that operate through employer businesses.

The nature of the adjustment shapes how its costs appear. A business closing is easy to count; a business that never opens is not. Enforcement suppresses formation without a corresponding rise in closures, so its costs accrue as foregone formation, a shortfall that cumulates as long as entry stays depressed yet may never register as a layoff or shutdown.

Evaluations centered on dislocation among existing firms can miss the margin that governs long-run local growth (Davis and Haltiwanger, 2014).

The findings connect a specific federal policy to the decline of U.S. business dynamism. Startup rates and the share of activity at young firms have fallen for decades (Haltiwanger et al., 2013; Decker et al., 2014, 2016), and slowing labor supply growth is a leading explanation, suppressing entry while leaving incumbent dynamics largely intact (Karahan et al., 2019; Hopenhayn et al., 2022). We document the same pattern from a discrete policy shock, placing interior immigration enforcement among the policy determinants of declining firm formation and access to immigrant labor among the forces that sustain it (Decker et al., 2020).

These findings are relevant to the current policy environment. The enforcement surge of 2025-26 operates at larger scale and through more direct instruments, including workplace raids, and early evidence shows it reducing work participation among undocumented immigrants with no offsetting gains for native workers (Cox and East, 2026). If the entry mechanism documented here carries over to this more intensive enforcement regime, part of its cost may surface as establishments that never appear, a margin that standard employment and wage statistics can miss.

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Figures

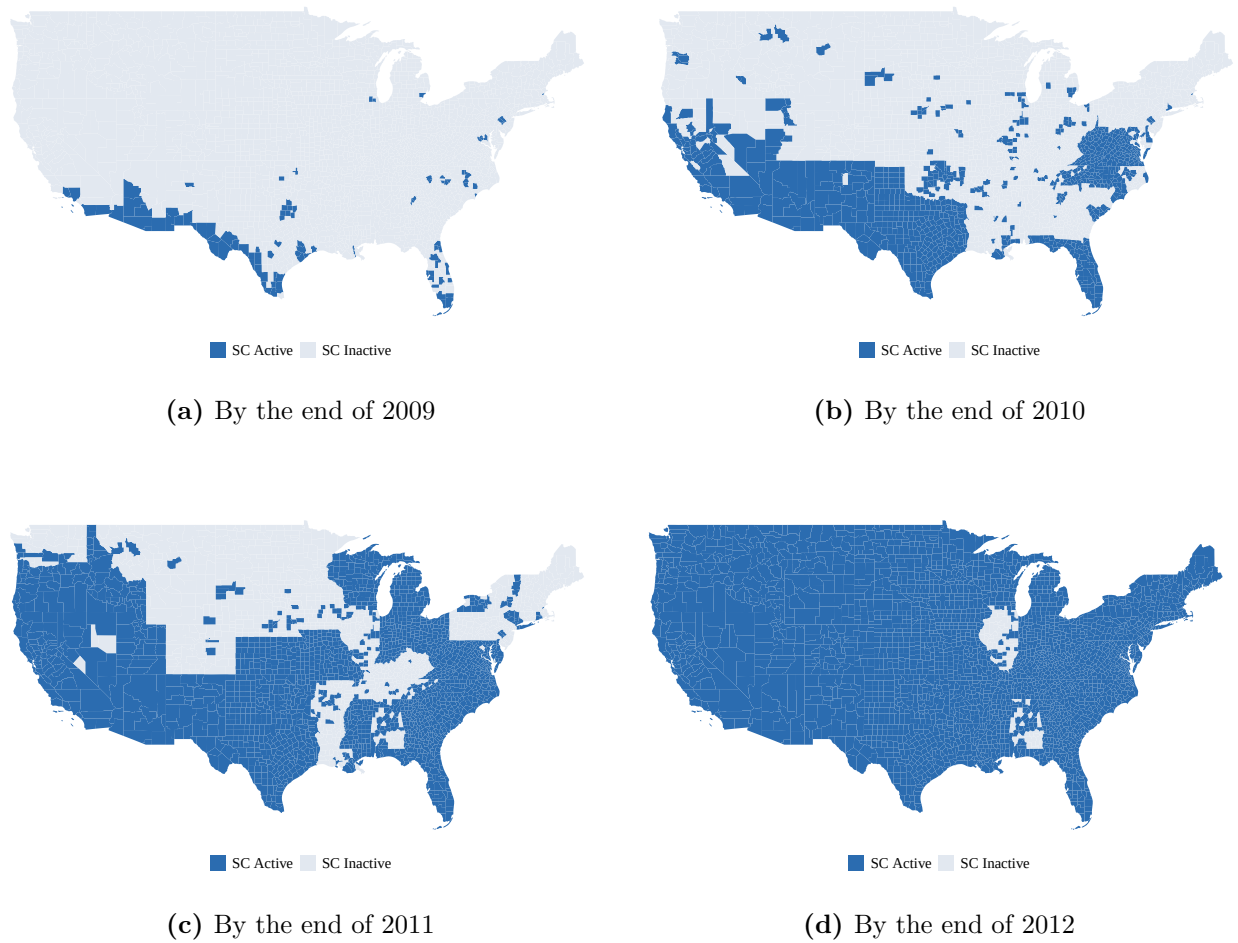
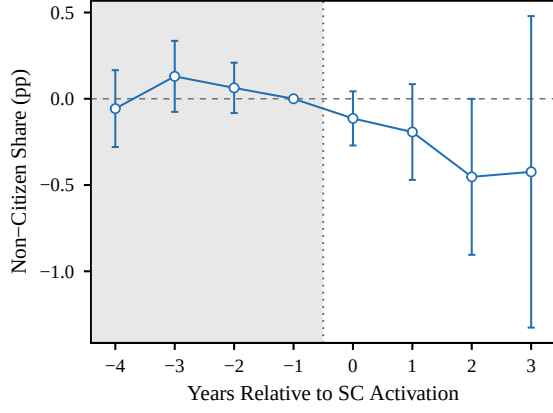
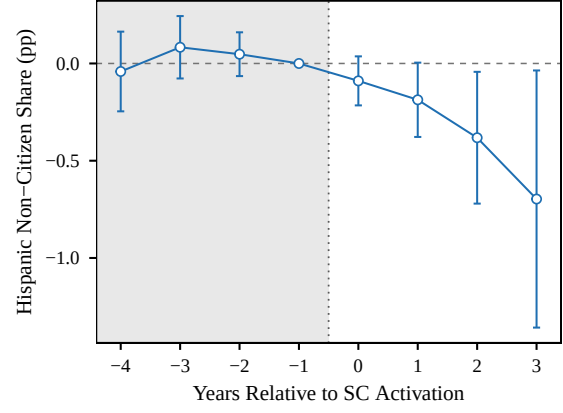


Figure 1: Secure Communities Rollout by County, 2009-2012

Notes. Each panel shows counties where Secure Communities had been activated by the end of the indicated calendar year. Coverage expanded rapidly over the rollout period, reaching approximately 64% of jurisdictions by 2011 and near-universal coverage (about 94%) by 2012.



(a) Non-Citizen Share



(b) Hispanic Non-Citizen Share

Figure 2: First Stage: Effect of SC on Immigrant Population Shares

Notes. Sun-Abraham event study estimates of SC activation on county-level population shares (ACS 1-year, counties with pop. $\geq 65,000$). Panel (a): non-citizen share; Panel (b): Hispanic non-citizen share. Event time $k = 0$ marks SC activation; $k = -1$ is the omitted reference period. County and year fixed effects with the full control set. Weighted by baseline county population. Standard errors clustered at the county level. Bars show 95% confidence intervals.

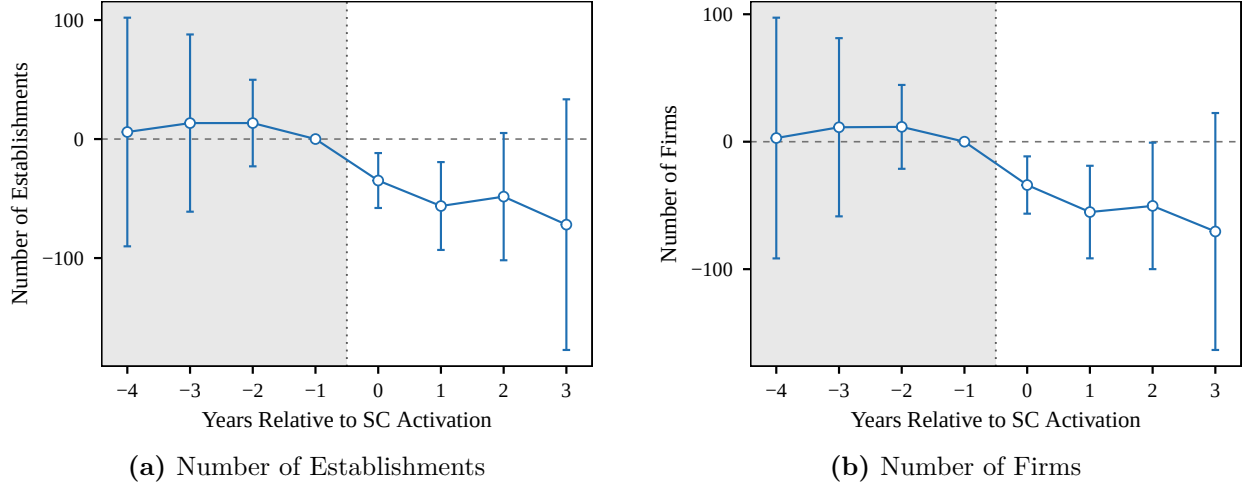


Figure 3: Event Study: Number of Firms and Establishments

Notes. [Sun and Abraham \(2021\)](#) interaction-weighted event study estimates. Panel (a) shows results for the total number of establishments and panel (b) for the total number of firms. Event time $k = 0$ marks the first year of Secure Communities activation in county c ; $k = -1$ is the omitted reference period. Controls include county-level and state-level enforcement indicators, the state minimum wage, a Bartik labor demand index, sanctuary policy status, and state-level housing boom exposure interacted with quadratic time trends. All specifications include county, year, and sector fixed effects, and are weighted by baseline county population. Standard errors clustered at the county level. Bars show 95% confidence intervals.

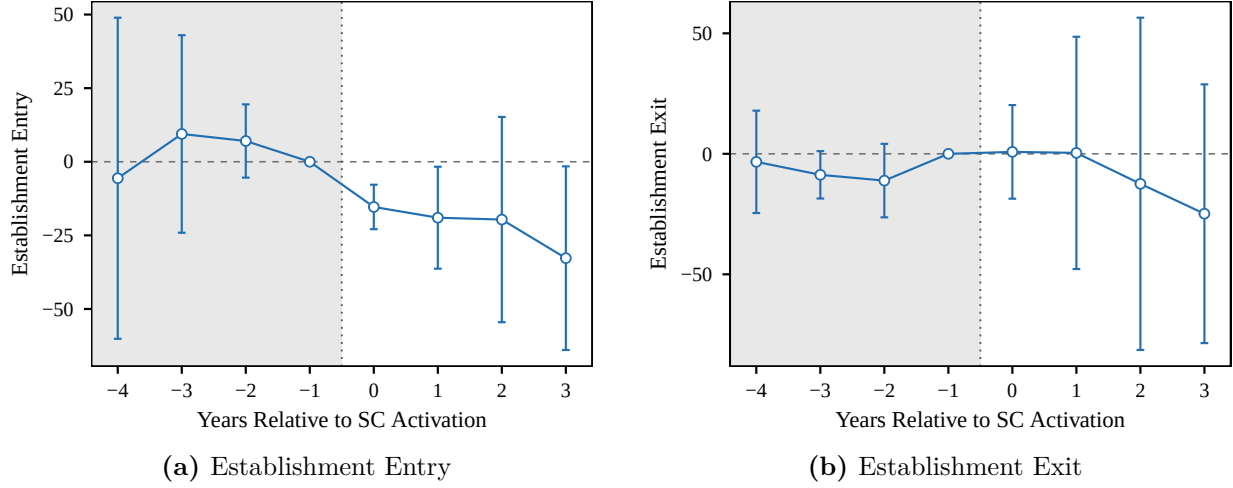


Figure 4: Event Study: Establishment Entry and Exit (Sun-Abraham)

Notes. Sun-Abraham cohort-specific event study estimates (Sun and Abraham, 2021). Estimation sample: 2005-2012; event window $[-4, +3]$. Panels show establishment entries and exits per county-sector-year cell. Event time $k = 0$ marks the first year of Secure Communities activation in county c ; $k = -1$ is the omitted reference period. Controls include county-level and state-level enforcement indicators, the state minimum wage, a Bartik labor demand index, sanctuary policy status, and state-level housing boom exposure interacted with quadratic time trends. All specifications include county, year, and sector fixed effects, and are weighted by baseline county population. Standard errors clustered at the county level. Bars show 95% confidence intervals.

Tables

Table 1: Effect of SC on the Number of Establishments and Firms

	Number of Establishments		Number of Firms	
	(1)	(2)	(3)	(4)
SC Activation	-34.4*	-45.7***	-35.8**	-45.2***
	(18.9)	(17.1)	(17.0)	(16.4)
Pre-treatment mean	1583.1	1583.1	1325.6	1325.6
% change	-2.2%	-2.9%	-2.7%	-3.4%
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Observations	410,707	410,707	410,707	410,707
R^2	0.661	0.661	0.651	0.651

Notes: Each column reports the [Sun and Abraham \(2021\)](#) aggregate ATT of Secure Communities activation on the indicated business count. Columns (1) and (3) omit the control set; columns (2) and (4) add it. The sample spans 2005-2012 at the county-sector-year level. Controls comprise county- and state-level enforcement indicators, the state minimum wage, a Bartik labor demand index, sanctuary policy status, and state-level housing boom exposure interacted with quadratic time trends. All specifications include county, year, and sector fixed effects, weighted by baseline county population (2000 Census). All columns are estimated on the same sample, restricted to county-years where every control is observed, so the columns differ only by the inclusion of the controls. The pre-treatment mean is constructed from each cohort's own pre-activation level and aggregated with the cohort weights of the Sun-Abraham estimator. Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Effects of Secure Communities on Establishment Entry and Exit

	Entry		Exit	
	(1)	(2)	(3)	(4)
SC Activation	-16.5*** (6.1)	-18.1** (7.7)	1.6 (22.9)	-2.8 (19.6)
Pre-treatment Mean	190.7	190.7	219.2	219.2
% Change	-8.7%	-9.5%	+0.7%	-1.3%
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Observations	298,867	298,867	301,151	301,151
R^2	0.673	0.674	0.677	0.678

Notes: [Sun and Abraham \(2021\)](#) aggregate ATT estimates. Dependent variables are establishment entries and exits per county-sector-year cell from the 2005-2012 BDS. Columns (1) and (3) omit the control set; columns (2) and (4) add it. Treatment cohort is defined by the first year SC is active for a majority of months in county c ; never-treated counties are the comparison group. All specifications include county, year, and sector fixed effects and are weighted by the 2000 Census baseline county population. Standard errors clustered at the county level. Observation counts vary across outcomes due to BDS cell-suppression thresholds. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 3: Heterogeneity by Immigrant Concentration

	Firms (1)	Establishments (2)	Estab. Entry (3)	Estab. Exit (4)
I. By sector immigrant share				
<i>Panel A: High immigrant-share sectors (10 sectors)</i>				
SC Activation	−70.1** (28.5) [−4.5%]	−71.1** (29.2) [−3.8%]	−28.5** (11.4) [−13.6%]	−0.6 (25.1) [−0.2%]
Observations	230,055	230,055	167,648	170,002
R^2	0.800	0.780	0.793	0.797
<i>Panel B: Low immigrant-share sectors (9 sectors)</i>				
SC Activation	−15.2 (10.2) [−1.4%]	−15.1 (10.6) [−1.2%]	−5.9 (7.3) [−3.5%]	−5.8 (13.8) [−3.2%]
Observations	180,652	180,652	131,219	131,149
R^2	0.514	0.543	0.560	0.558
II. By county immigrant share				
<i>Panel C: High immigrant-share counties (1,551 counties)</i>				
SC Activation	−52.4** (20.9) [−3.4%]	−53.6** (22.1) [−2.9%]	−19.9** (8.1) [−9.2%]	−2.2 (22.8) [−0.9%]
Observations	207,999	207,999	158,494	159,377
R^2	0.662	0.672	0.683	0.686
<i>Panel D: Low immigrant-share counties (1,546 counties)</i>				
SC Activation	0.1 (0.5) [+0.1%]	0.3 (0.5) [+0.2%]	0.7*** (0.2) [+5.8%]	−1.5*** (0.4) [−9.9%]
Observations	202,708	202,708	140,373	141,774
R^2	0.679	0.676	0.663	0.661
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Sun and Abraham (2021) aggregate ATT using 2005-2012 BDS data for the number of firms, the number of establishments, establishment entry, and establishment exit. Panels A-B split sectors at the median foreign-born low-education employment share (6.3%, ACS 1-year PUMS, 2005-2007). Panels C-D split counties at the median baseline foreign-born share. Percentage changes relative to pre-treatment treated-county means appear in square brackets. All specifications include county, year, and sector fixed effects with the full control set and are weighted by baseline county population. Standard errors clustered at the county level. Separate-sample estimates do not constitute formal tests of cross-group differences. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Heterogeneity by Sector Baseline Turnover Rate

	Firms (1)	Establishments (2)	Estab. Entry (3)	Estab. Exit (4)
<i>Panel A: High baseline turnover sectors (above median, 10 sectors)</i>				
SC Activation	−63.4*** (23.9) [−4.0%]	−64.8** (26.0) [−3.4%]	−26.6** (11.8) [−11.3%]	3.6 (28.5) [+1.3%]
Observations	220,277	220,277	163,757	165,502
R^2	0.703	0.700	0.720	0.723
<i>Panel B: Low baseline turnover sectors (below median, 9 sectors)</i>				
SC Activation	−18.6* (10.9) [−1.8%]	−19.4* (10.9) [−1.5%]	−9.5* (5.4) [−6.8%]	−5.7 (13.1) [−3.6%]
Observations	190,430	190,430	135,110	135,649
R^2	0.596	0.625	0.650	0.654
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: [Sun and Abraham \(2021\)](#) aggregate ATT for the number of firms, the number of establishments, establishment entry, and establishment exit, estimated separately on high- and low-turnover sector subsamples using 2005-2012 BDS data. Sectors split at the median baseline turnover rate (9.7%, QWI 2005-2007 hire-weighted average). Turnover is the average quarterly ratio of hires plus separations to employment. Percentage changes relative to pre-treatment treated-county means appear in brackets. All specifications include county, year, and sector fixed effects with the full control set. Weighted by baseline county population. Standard errors clustered at the county level. Separate-sample estimates do not constitute formal tests of cross-group differences. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Heterogeneity by Tradable vs. Non-Tradable Sectors

	Firms (1)	Establishments (2)	Estab. Entry (3)	Estab. Exit (4)
<i>Panel A: All sectors (pooled benchmark)</i>				
SC Activation	−45.2*** (16.4) [−3.4%]	−45.7*** (17.1) [−2.9%]	−18.1** (7.7) [−9.5%]	−2.8 (19.6) [−1.3%]
Observations	410,707	410,707	298,867	301,151
<i>Panel B: Tradable sectors</i>				
SC Activation	−1.4 (24.5) [−0.3%]	−2.3 (24.9) [−0.4%]	−7.5 (6.1) [−14.4%]	12.8 (20.0) [+15.8%]
Observations	57,152	57,152	39,582	40,040
<i>Panel C: Nontradable sectors</i>				
SC Activation	−58.8*** (20.7) [−4.0%]	−58.3*** (21.6) [−3.3%]	−17.8** (8.2) [−8.4%]	−9.1 (17.6) [−3.8%]
Observations	353,554	353,554	259,281	261,106
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Sun and Abraham (2021) aggregate ATT for the number of firms, the number of establishments, establishment entry, and establishment exit, using 2005-2012 BDS data. Panel A pools all sectors and reproduces the aggregate estimates reported earlier; Panels B and C re-estimate the same specification on the tradable and nontradable subsamples. Tradable sectors are agriculture (11), mining (21), and manufacturing (31-33), following Burstein et al. (2020); all remaining sectors, including construction, wholesale, retail, and services, are nontradable. Table A.8 reports the immigrant employment share of each bin. Percentage changes relative to pre-treatment treated-county means appear in brackets; because the pre-treatment means differ across panels, the bracketed percentages rather than the level coefficients are the comparable quantities. The difference between Panels B and C is not reported here, since combining separately estimated standard errors would assume the two subsamples are independent when they share counties; Table A.5 tests it with a county cluster bootstrap. All specifications include county, year, and sector fixed effects with the full control set. Weighted by baseline county population. Standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Effects of Secure Communities on Job Creation and Destruction

	(1) Job Creation	(2) Job Destruction
SC Activation	-394.4** (186.5)	109.7 (594.0)
Pre-treatment Mean	3,843.2	4,923.9
% Change	-10.3%	+2.2%
County Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Sector Fixed Effects	Yes	Yes
Controls	Yes	Yes
Observations	410,707	408,373
R^2	0.671	0.663

Notes: [Sun and Abraham \(2021\)](#) aggregate ATT estimates. Dependent variables are job creation and job destruction per county-sector-year cell from the 2005-2012 BDS. Treatment cohort defined by first year SC is active for a majority of months in county c ; never-treated counties are the comparison group. All specifications include county, year, and sector fixed effects with the full control set, and are weighted by the 2000 Census baseline county population. Standard errors clustered at the county level. Observation counts vary across outcomes due to BDS cell-suppression thresholds. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 7: Job Flow Decomposition: Births vs. Continuers

	Job Creation		Job Destruction	
	Estab. Births (1)	Continuers (2)	Estab. Deaths (3)	Continuers (4)
SC Activation	-61.3 (45.3)	-354.5** (171.0)	-19.3 (169.4)	164.6 (476.0)
Pre-treatment Mean	1,529	2,484	1,689	3,420
% Change	-4.0%	-14.3%	-1.1%	+4.8%
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	298,867	404,201	301,151	404,201
R^2	0.012	0.008	0.007	0.009

Notes: [Sun and Abraham \(2021\)](#) aggregate ATTs for the 2005-2012 BDS job flow decomposition. Job creation is decomposed into jobs created at establishments born in year t (births) and jobs created at establishments that existed in both $t - 1$ and t (continuers). Job destruction is decomposed analogously into deaths and continuers. All specifications include county, year, and sector fixed effects with the full control set. Weighted by baseline county population. Standard errors clustered at the county level. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 8: Effect of SC Activation on Incorporated Self-Employment

	Non-Citizen		U.S. Citizen		Total		Nonemployer
	Rate (1)	Count (2)	Rate (3)	Count (4)	Rate (5)	Count (6)	Count (7)
SC Activation	0.206 (0.183)	545 (335)	-0.085 (0.065)	-10 (536)	-0.040 (0.066)	535 (802)	-2,039 (2,855)
Pre-treatment Mean	2.440	2,374	3.521	17,792	3.385	20,166	104,605
% Change	+8.4%	+23.0%	-2.4%	-0.1%	-1.2%	+2.7%	-1.9%
Observations	24,303	24,748	24,748	24,748	24,748	24,748	24,760
R^2	0.180	0.977	0.793	0.997	0.782	0.996	0.998
County Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: [Sun and Abraham \(2021\)](#) aggregate ATT estimates. In columns (1)-(6) the county-year panel is constructed from official 2005-2012 ACS 1-year PUMS files and PUMA-to-county population crosswalks. Incorporated self-employment is the ACS class-of-worker category for self-employed workers whose businesses are incorporated. The rate is the percentage of employed workers in that category and is measured in percentage points; the number is the corresponding person-weighted count. Columns (1)-(2) report non-citizens, columns (3)-(4) U.S. citizens, and columns (5)-(6) all workers. Column (7) reports the annual number of all-industry nonemployer establishments per county from the 2005-2012 Census Nonemployer Statistics; nonemployer establishments are businesses with no paid employees, and the source does not identify owners by citizenship. All specifications include county and year fixed effects with the full control set and are weighted by baseline county population. Standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

ONLINE SUPPLEMENTAL APPENDIX

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A Additional Figures and Tables

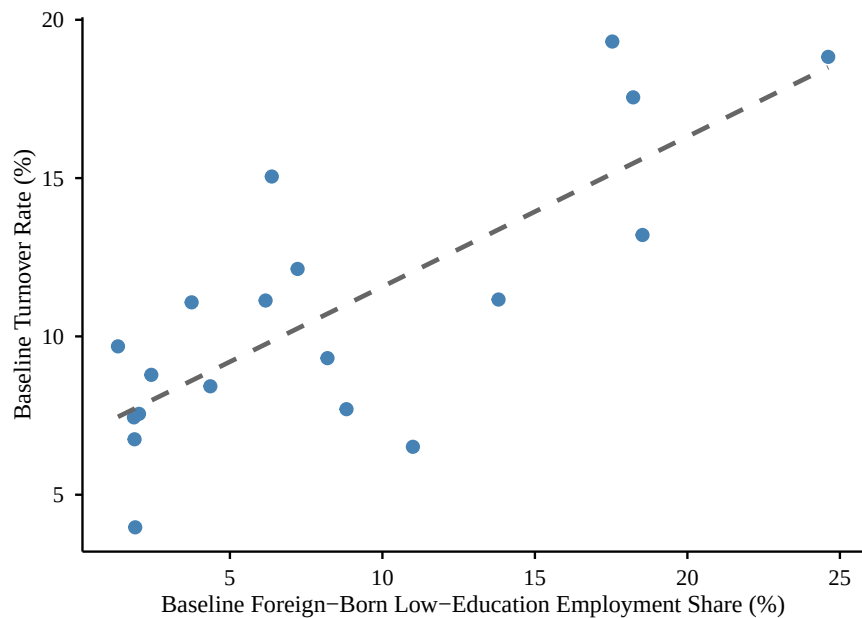
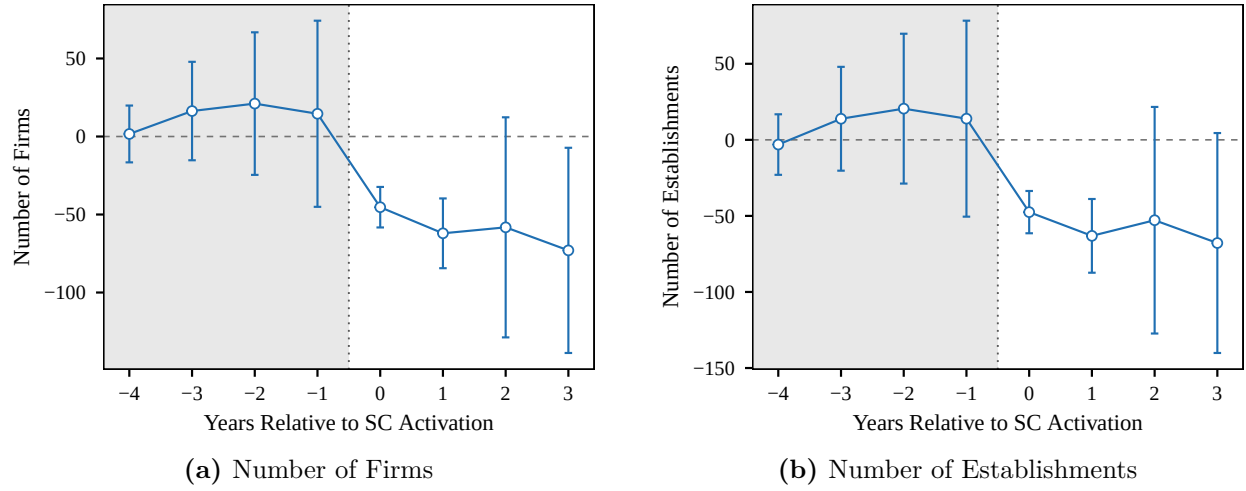


Figure A.1: Sector Immigrant Share and Baseline Turnover Rate

Notes. Each point is a 2-digit NAICS sector. The x-axis is the national share of employed workers who are foreign-born with at most a high school education (ACS 1-year PUMS, 2005-2007). The y-axis is the baseline turnover rate (average quarterly ratio of hires plus separations to employment, QWI 2005-2007, hire-weighted). Dashed line is OLS fit. Pearson correlation $r = 0.77$.

Figure A.2: Event Study: Borusyak-Jaravel-Spiess Imputation Estimator



Notes: Event-study coefficients from the [Borusyak et al. \(2024\)](#) imputation estimator. Each panel plots the dynamic treatment effect of SC activation on the indicated outcome, with 95% confidence intervals. Counterfactual outcomes for treated units are imputed from untreated observations using county-sector and year fixed effects and the full control set. County-sector-year panel, 2005-2012. Weighted by baseline county population. Standard errors clustered at the county level.

Table A.1: County Counts by Treatment Group, 2005-2012

	Counties
SC activated in 2009	51
SC activated in 2010	276
SC activated in 2011	1,026
SC activated in 2012	1,627
Never-treated (not activated by 2012)	176
<i>Total counties</i>	<i>3,156</i>

Notes: Panel A tabulates counties by first-activation year of Secure Communities; “Never-treated” denotes counties whose first activation falls after 2012. Panel B reports the count of counties ever exposed to each enforcement policy during 2005-2012. Policy indicators are coded 1 if active in a majority of months in a given year. Categories in Panel B are not mutually exclusive. Data: enforcement panel used in all main specifications.

Table A.2: Descriptive Statistics

	Unweighted		Weighted	
	Mean	SD	Mean	SD
<i>Panel A: Business Dynamics (County Level)</i>				
No. of Firms	1,879	5,986	20,837	36,218
No. of Establishments	2,139	7,035	24,507	42,564
Establishment Entry	248	916	3,125	5,643
Establishment Exit	210	775	2,653	4,746
Job Creation	5,650	21,371	71,708	126,014
Job Destruction	5,094	19,417	66,087	117,802
Observations	3,147			
<i>Panel B: Business Dynamics (County-Sector Level)</i>				
No. of Firms	109	440	1,122	2,618
No. of Establishments	124	512	1,320	3,053
Establishment Entry	16	69	174	403
Establishment Exit	14	58	148	337
Job Creation	328	1,630	3,862	8,649
Job Destruction	298	1,401	3,564	8,134
Observations	59,793			
<i>Panel C: Demographic Characteristics</i>				
% Foreign-Born Non-Citizen	4.46	4.34	8.09	6.07
% Foreign-Born, $\leq$ HS Education	3.32	4.99	8.57	8.26
% Hispanic Non-Citizen	3.45	3.66	5.18	4.55
Observations	3,147			

Notes. Weighted columns show the mean and standard deviation of the variables weighted by baseline county population. Panels A and B report pre-treatment averages (2005-2007) from the Business Dynamics Statistics (BDS), rounded to integers. Panel A aggregates across sectors within each county-year, then averages over the three years. Panel B reports county-sector-level averages. Panel C reports county-level demographic shares from the American Community Survey (ACS): % Foreign-Born Non-Citizen from ACS 1-year estimates (2005-2007); % Foreign-Born with $\leq$ HS Education from ACS 5-year estimates (2006-2010); % Hispanic Non-Citizen from ACS 1-year estimates (2005-2007).

Table A.3: Pre-Period Covariate Balance: Predictors of SC Adoption Timing

	First Treatment Year
Δ Establishments	−0.000** (0.000)
Δ Share Non-Citizen	−0.019 (0.026)
Δ Share Non-Hispanic White	0.013 (0.021)
Δ Share Unemployed (LF)	−0.008 (0.011)
Δ Log Housing Price Index	−0.159 (0.869)
Republican Vote Share	−0.455 (0.573)
Observations	2,690
R^2	0.167

Notes: OLS regression of first SC adoption year on 2005-2007 changes in county-level pre-period characteristics. All Δ variables reflect changes between 2005 and 2007. Share variables are expressed in percentage points. Regression weighted by 2000 Census baseline county population. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: First Stage: Effect of SC on Immigrant Population Shares

	Non-Citizen Share (1)	Foreign-Born Share (2)	Hispanic Non-Citizen (3)
SC Activation	-0.218 (0.133)	-0.453*** (0.144)	-0.211** (0.099)
Pre-treatment mean	8.637	14.991	5.888
% change	-2.5%	-3.0%	-3.6%
County Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	6,118	6,223	3,836
R^2	0.991	0.997	0.989

Notes: [Sun and Abraham \(2021\)](#) aggregate ATT. County-year panel, 2005-2012, ACS 1-year estimates for counties with population $\geq 65,000$. Non-citizen share: non-citizen population / total population. Foreign-born share: foreign-born population / total population. Hispanic non-citizen share: Hispanic non-citizen population / total population. All specifications include county and year fixed effects with the full control set. Weighted by baseline county population. Standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Formal Tests of Differences Across Exposure Subgroups

	Firms (1)	Establishments (2)	Estab. Entry (3)	Estab. Exit (4)
Nontradable – Tradable	−57.4 [38.4]	−56.0 [40.5]	−10.2 [8.5]	−21.9*** [7.2]
Imm.-intensive – Not imm.-intensive	−54.9** [26.4]	−56.0** [28.0]	−22.6* [13.3]	5.2 [13.0]
High turnover – Low turnover	−44.8* [24.1]	−45.4* [27.4]	−17.1 [11.8]	9.3 [18.8]
Bootstrap replications	300			

Notes: Each row reports the difference between the [Sun and Abraham \(2021\)](#) aggregate ATTs of the two subgroups indicated, for the four business outcomes. The Sun-Abraham estimator aggregates cohort-by-time ATTs and does not admit an interaction term, and the sector subsamples are drawn from the same counties, so combining their separately estimated standard errors under an independence assumption is invalid. Each difference is therefore tested by resampling counties with replacement and re-estimating *both* subgroup ATTs on the same resample, so that the bootstrap distribution of the difference carries the covariance between them. Bootstrap standard errors appear in brackets; p -values use a normal approximation. Immigrant intensity and turnover split sectors at the median; tradability follows [Burstein et al. \(2020\)](#). Stars denote significance of the *difference*, not of either subgroup estimate. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Sectors by Baseline Immigrant Share

NAICS	Sector	Immigrant Share
<i>Panel A: Above-Median Immigrant Share</i>		
11	Agriculture, Forestry, Fishing & Hunting	0.246
23	Construction	0.185
72	Accommodation & Food Services	0.182
56	Admin., Support & Waste Mgmt.	0.175
81	Other Services	0.138
31-33	Manufacturing	0.110
42	Wholesale Trade	0.088
48-49	Transportation & Warehousing	0.082
44-45	Retail Trade	0.072
71	Arts, Entertainment & Recreation	0.064
<i>Panel B: Below-Median Immigrant Share</i>		
53	Real Estate & Rental/Leasing	0.062
62	Health Care & Social Assistance	0.044
21	Mining, Quarrying & Oil/Gas	0.037
51	Information	0.024
52	Finance & Insurance	0.020
22	Utilities	0.019
61	Educational Services	0.019
55	Management of Companies	0.019
54	Professional & Technical Services	0.013

Notes. Immigrant share is the fraction of sector employment that is foreign-born with at most a high school education, from the American Community Survey (ACS) 1-year PUMS, 2005-2007. Sectors are split at the median immigrant share across sectors.

Table A.7: Sectors by Baseline Turnover Rate

NAICS	Sector	Turnover Rate
<i>Panel A: Above-Median Turnover</i>		
56	Admin., Support & Waste Mgmt.	0.193
11	Agriculture, Forestry, Fishing & Hunting	0.188
72	Accommodation & Food Services	0.176
71	Arts, Entertainment & Recreation	0.150
23	Construction	0.132
44-45	Retail Trade	0.121
81	Other Services	0.112
53	Real Estate & Rental/Leasing	0.111
21	Mining, Quarrying & Oil/Gas	0.111
54	Professional & Technical Services	0.097
<i>Panel B: Below-Median Turnover</i>		
48-49	Transportation & Warehousing	0.093
51	Information	0.088
62	Health Care & Social Assistance	0.084
42	Wholesale Trade	0.077
52	Finance & Insurance	0.076
55	Management of Companies	0.074
61	Educational Services	0.068
31-33	Manufacturing	0.065
22	Utilities	0.040

Notes. Turnover rate is the hire-weighted national average of the quarterly ratio of hires plus separations to employment, from the Quarterly Workforce Indicators (QWI), 2005-2007. Sectors are split at the median turnover rate across sectors.

Table A.8: Immigrant Employment Share of the Tradable and Nontradable Bins

	Sectors (1)	LEFB Share (%) (2)
<i>Burstein et al. (2020)</i>		
Tradable	3	11.8
Nontradable	17	7.3
<i>Legacy composite</i>		
Tradable	7	7.7
Nontradable	13	8.0

Notes: Column 1 reports the number of two-digit NAICS sectors assigned to each bin; column 2 reports the share of each bin's employment accounted for by low-educated foreign-born workers, computed from the 2005-2007 ACS and aggregated to BDS sector codes. Under the [Burstein et al. \(2020\)](#) classification used in Table 5, the tradable bin is the more immigrant-intensive of the two, so a labor supply contraction operating alone would predict larger declines in tradable sectors.

Table A.9: Job Creation Decomposition by Sector Turnover

	Job Creation (Births) (1)	Job Creation (Continuers) (2)
<i>High turnover sectors</i>		
SC Activation	−220.5** (107.2) [−12.4%]	−418.7** (199.4) [−16.2%]
Pre-treatment mean	1,783	2,591
<i>Low turnover sectors</i>		
SC Activation	115.5** (55.8) [+9.3%]	−289.6* (150.2) [−12.2%]
Pre-treatment mean	1,245	2,367
Observations	298,867	404,201
County Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Sector Fixed Effects	Yes	Yes
Controls	Yes	Yes

Notes: [Sun and Abraham \(2021\)](#) aggregate ATT for job creation at establishment births and at continuing establishments, estimated separately on sectors above and below the median baseline turnover rate. Turnover is measured from 2005-2007 QWI data and split at the median, the same construction used in Table 4. Percentage changes relative to pre-treatment treated-county means appear in brackets. All specifications include county, year, and sector fixed effects with the full control set, are weighted by baseline county population, and cluster standard errors by county. These are separate-subsample estimates; the cross-group difference is not formally tested here. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10: Spatial Robustness: Commuting Zone Tests

	Firms (1)	Establishments (2)	Estab. Entry (3)	Estab. Exit (4)
<i>Panel A: County + Year + Sector FE (baseline)</i>				
SC Activation	-45.2*** (16.4)	-45.7*** (17.1)	-18.1** (7.7)	-2.8 (19.6)
Pre-treatment mean	1,325.6	1,583.1	190.7	219.2
% change	-3.4%	-2.9%	-9.5%	-1.3%
Observations	410,707	410,707	298,867	301,151
R^2	0.651	0.661	0.674	0.678
<i>Panel B: County + CZ $\times$ Year + Sector FE</i>				
SC Activation	-29.2** (11.8)	-28.0** (12.3)	-17.8*** (5.1)	10.2 (10.0)
Pre-treatment mean	1,325.6	1,583.1	190.7	219.2
% change	-2.2%	-1.8%	-9.4%	+4.6%
Observations	410,707	410,707	298,867	301,151
R^2	0.652	0.661	0.677	0.680
<i>Panel C: CZ + Year + Sector FE (commuting-zone level)</i>				
SC Activation	-176.5** (69.8)	-197.7** (78.5)	-57.0*** (20.6)	18.6 (31.3)
Pre-treatment mean	3,347.7	3,943.5	411.3	484.1
% change	-5.3%	-5.0%	-13.9%	+3.9%
Observations	109,592	109,592	109,592	109,592
R^2	0.671	0.680	0.674	0.679
Controls	Yes	Yes	Yes	Yes

Notes: Panel A reproduces the baseline Sun-Abraham aggregate ATTs with county, year, and sector fixed effects. Panel B replaces year fixed effects with commuting zone $\times$ year fixed effects, absorbing all time-varying confounds at the labor market level. Panel C aggregates the panel to the commuting zone $\times$ sector $\times$ year level and re-estimates with CZ, year, and sector fixed effects; the treatment cohort is the year when the cumulative population-weighted share of constituent counties under SC exceeds 50 percent; CZ clustering. Commuting zones follow [Tolbert and Sizer \(1996\)](#). All specifications include the full control set, weighted by baseline population. Standard errors clustered at the county level in Panels A-B, at the CZ level in Panel C. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.11: Missing Values in Outcome Variables

	Total Panel	Non-missing	Missing	% Missing
Number of firms	479,712	417,803	61,909	12.91
Number of establishments	479,712	417,803	61,909	12.91
Establishment entry	479,712	304,709	175,003	36.48
Establishment exit	479,712	306,697	173,015	36.07

Notes: The table reports the total number of observations, non-missing observations, missing observations, and the percentage of missing observations for each outcome variable in the county-sector-year panel.

Table A.12: Balanced Panel Estimates

	Number of Firms	Number of Establishments	Establishment Entry	Establishment Exit
ATT	-52.0*** (19.2)	-51.8*** (19.9)	-17.9** (7.8)	-8.4 (18.4)
Pre-treatment mean	1350.4	1612.8	225.3	260.8
% change	-3.9%	-3.2%	-7.9%	-3.2%
Observations	378,192	378,192	138,624	140,272
R^2	0.654	0.664	0.715	0.720
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Sun and Abraham (2021) aggregate ATTs for the number of firms, the number of establishments, establishment entry, and establishment exit, restricted to county-sector cells present in every sample year. Outcome-specific BDS suppression can still produce missing values within this structural panel. All specifications use the full control set and are weighted by baseline county population. Standard errors clustered at the county level. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table A.13: Missing Data Selection Test

	Number of Firms	Number of Establishments	Establishment Entry	Establishment Exit
ATT	0.0084 (0.0056)	0.0084 (0.0056)	-0.0099 (0.0090)	0.0213** (0.0096)
Baseline missing rate	12.9%	12.9%	36.5%	36.1%
Observations	470,440	470,440	470,440	470,440
R^2	0.181	0.181	0.312	0.308
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: The dependent variable is an indicator for whether the county-sector-year outcome is suppressed in the BDS. A significant positive coefficient indicates that SC activation increases the probability of suppression; it does not identify the direction of selection in the unobserved outcome values. All specifications use the full control set and are weighted by baseline county population. Standard errors clustered at the county level. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table A.14: Robustness: Trimming Bounds

	Number of Firms	Number of Establishments	Establishment Entry	Establishment Exit
<i>Panel A: Trim bottom 1%</i>				
ATT	-47.7*** (17.1) [-3.6%]	-47.5*** (18.1) [-3.0%]	-18.2** (8.0) [-9.4%]	-2.7 (19.7) [-1.2%]
Observations	375,822	375,822	265,466	269,199
R^2	0.653	0.663	0.676	0.680
<i>Panel B: Trim bottom 5%</i>				
ATT	-58.5*** (18.5) [-4.3%]	-59.4*** (20.8) [-3.7%]	-17.8** (8.0) [-9.2%]	-2.5 (19.8) [-1.1%]
Observations	361,272	361,272	256,804	259,266
R^2	0.654	0.664	0.677	0.681
<i>Panel C: Trim bottom 10%</i>				
ATT	-63.5*** (17.4) [-4.7%]	-66.0*** (17.5) [-4.1%]	-15.9** (7.4) [-8.2%]	-4.2 (19.3) [-1.9%]
Observations	341,473	341,473	244,985	247,102
R^2	0.656	0.666	0.679	0.683
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: [Sun and Abraham \(2021\)](#) aggregate ATTs after trimming county-sector cells with the smallest baseline establishment counts. Panel A: bottom 1% trimmed. Panel B: bottom 5 percent. Panel C: bottom 10 percent. Percentage changes in brackets. All specifications include county, year, and sector fixed effects, weighted by baseline county population. Standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.15: Borusyak-Jaravel-Spiess Imputation Estimates

	Firms (1)	Establishments (2)
BJS ATT	-54.1*** (13.2)	-54.4*** (14.1)
Pre-treatment Mean	1,326	1,583
% Change	-4.1%	-3.4%
Observations	417,803	417,803
R^2	0.999	0.999

Notes: Static ATTs from the [Borusyak et al. \(2024\)](#) imputation estimator. Counterfactual outcomes for treated units are imputed from untreated observations using county-sector and year fixed effects and the full control set. Dynamic coefficients shown in Figure A.2 come from separate event-study estimations. County-sector-year panel, 2005-2012. Weighted by baseline county population. Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.16: Robustness: Year FE with Controls vs. State $\times$ Year FE

	Firms (1)	Establishments (2)	Estab. Entry (3)	Estab. Exit (4)
<i>Panel A: Sun-Abraham – Year FE (primary)</i>				
ATT	-45.2*** (16.4)	-45.7*** (17.1)	-18.1** (7.7)	-2.8 (19.6)
<i>Panel B: Sun-Abraham – State $\times$ Year FE (robustness)</i>				
ATT	-35.6*** (12.4)	-37.9*** (13.0)	-17.9*** (5.2)	6.5 (12.8)
<i>Panel C: TWFE – Year FE (primary)</i>				
ATT	-30.2** (15.2)	-30.1** (14.7)	-17.0** (7.1)	0.1 (4.0)
<i>Panel D: TWFE – State $\times$ Year FE (robustness)</i>				
ATT	-17.6** (7.3)	-17.9** (7.3)	-10.8*** (3.1)	2.6 (2.6)
Observations	479,712			

Notes: Panels A and B report Sun-Abraham aggregate ATTs using county, sector, and year fixed effects, and county, sector, and state-by-year fixed effects, respectively, along with a full set of control variables. Weighted by baseline county population. Standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Immigration Enforcement Control Variables

287(g) County. Section 287(g) of the Immigration and Nationality Act, added by IIRIRA in 1996, authorizes formal agreements between ICE and state and local law enforcement agencies allowing designated officers to perform specified immigration enforcement functions under federal supervision. County-level 287(g) agreements authorize designated local officers, often in jail settings and in some periods or jurisdictions in patrol or task-force roles, to identify and process removable immigrants encountered during routine law enforcement operations. We include an indicator for whether a county had an active 287(g) agreement in a given year to absorb enforcement variation operating through this channel, which predates and partially overlaps with the SC rollout.

287(g) State. In addition to county-level agreements, some states entered into 287(g) agreements at the state agency level, typically through state police or corrections departments. These agreements extend immigration enforcement authority across broader jurisdictions than county-level agreements and may affect counties that did not independently adopt 287(g) programs. We include a separate indicator for state-level 287(g) agreements to capture this additional layer of enforcement exposure that is not absorbed by the county-level indicator.

E-Verify. E-Verify is an internet-based employment eligibility verification system administered by the Department of Homeland Security that allows employers to confirm the work authorization status of newly hired employees by checking their information against federal databases. Unlike SC, which operates through the criminal justice system, E-Verify targets the employment relationship directly, raising the cost of hiring undocumented workers for participating employers. Several states mandated E-Verify participation for some or all employers during our sample period. We include an indicator for whether a state E-Verify mandate was active in a given year to ensure that our SC estimates are not confounded by this distinct employer-facing enforcement mechanism.

Omnibus Immigration Laws. Several states enacted comprehensive omnibus immigration legislation during our sample period that bundled multiple enforcement and restriction measures into single legislative acts. These laws varied in scope but often included provisions related to employment verification, eligibility verification for certain state and local public benefits, access to driver's licenses and educational benefits, and expanded local law enforcement authority to enforce immigration laws. Because omnibus laws alter the broader environment facing undocumented immigrants and may affect both labor supply and local

demand conditions, we include an indicator for whether a state omnibus immigration law was active in a given year.

Sanctuary Policy. Sanctuary policies generally refer to formal policies adopted by city, county, or state governments that limit the cooperation of local law enforcement agencies with federal immigration enforcement authorities. These policies typically take the form of directives prohibiting local officers from inquiring about immigration status, declining to honor ICE detainer requests, or restricting information sharing with federal immigration authorities. Because sanctuary policies may partially offset the enforcement effects of SC by reducing the probability that local criminal justice contact leads to federal immigration consequences, we control for sanctuary policy status to isolate the effect of SC activation. For each county, we identify the earliest adoption date of any sanctuary policy at the city, county, or state level and define annual treatment as an indicator equal to one if a policy is in effect for at least six months of a given year.

C Construction of the Bartik Labor Demand Index

We construct a Bartik shift-share labor demand index following [Bartik \(1991\)](#) to control for differential local demand shocks that might confound the effect of immigration enforcement on business dynamics. The index interacts baseline county industry composition with national sectoral employment growth to generate predicted county-level labor demand, isolating demand variation that is plausibly exogenous to local enforcement decisions ([Autor et al., 2013](#); [Goldsmith-Pinkham et al., 2020](#)).

Let $\text{emp}_{c,s,0}$ denote employment in county c , sector s in the base year (2001), and let $\text{emp}_{c,0} = \sum_s \text{emp}_{c,s,0}$ denote total county employment. The baseline industry share is

$$\omega_{c,s} = \frac{\text{emp}_{c,s,0}}{\text{emp}_{c,0}}. \quad (5)$$

We compute leave-one-out national sector employment growth to avoid mechanical correlation between the county’s own employment and the national growth rate. Define $N_{-c,s,t} = \sum_{c' \neq c} \text{emp}_{c',s,t}$ as total national employment in sector s in year t , excluding county c . The cumulative growth ratio is

$$g_{-c,s,t} = \frac{N_{-c,s,t}}{N_{-c,s,0}}. \quad (6)$$

The Bartik index for county c in year t is then

$$B_{c,t} = \sum_s \omega_{c,s} \cdot g_{-c,s,t}. \quad (7)$$

Values near one indicate that predicted labor demand in county c is close to its 2001 level; values above one indicate positive predicted demand growth. Employment data for both the baseline shares and the national growth rates come from the BDS. We use this index as a time-varying control in all regressions to absorb local labor demand shocks unrelated to enforcement.

D Sample Window and Cohort Composition

Our sample ends in 2012. Two questions follow: why not extend it, and what identifies the longer-horizon event-time coefficients within it.

Why the window ends in 2012. Two considerations drive this restriction. First, the [Sun and Abraham \(2021\)](#) estimator, like [Callaway and Sant’Anna \(2021\)](#), identifies treatment effects by comparing treated units with units that are not yet treated or never treated, and that pool is nearly exhausted by the end of our window. Table [D.1](#) shows that 2,980 of 3,156 counties, or 94.4%, have activated SC by 2012, leaving 176 never-treated counties. Extending the sample past 2012 would therefore add post-treatment observations for already-treated counties while adding almost nothing to the set of units that identify the counterfactual, eroding the comparison group the design depends on rather than lengthening the horizon over which effects are credibly estimated. Second, SC was replaced nationally by the Priority Enforcement Program (PEP) in 2014, and because this transition occurred simultaneously across counties rather than through a staggered rollout, a PEP indicator would be close to collinear with year fixed effects, absorbing national time variation rather than isolating a distinct policy.

What identifies the longer horizons. A cohort activated in year g can contribute to event time k only if $g + k \leq 2012$. Later event times are therefore mechanically populated by earlier cohorts. Table [D.1](#) makes this concrete. The contemporaneous coefficient $k = 0$ draws on all four cohorts and 2,980 treated counties. By $k = +2$ only the 2009 and 2010 cohorts remain, covering 327 counties, and $k = +3$ is identified by the 2009 cohort alone, covering 51 counties.

Composition shifts along with sample size. Counties in border or southern states make up 50% of those contributing at $k = 0$ but 91.1% at $k = +2$ and 94.1% at $k = +3$. The early cohorts are geographically distinctive: the 2009 cohort is concentrated in Texas, North Carolina, Florida, and Arizona, while the never-treated group is only 31.2% border or southern. The long-horizon coefficients thus describe a specific and unrepresentative set of early-activating counties rather than the average treated county.

This also limits the robustness check one would naturally want, namely re-estimating the long horizons after excluding border and southern counties. At $k = +3$ that exclusion would leave roughly three counties, so the test cannot be run at the horizons where it would matter. We regard this as informative in itself: the data cannot distinguish genuine persistence from the particular trajectory of early-activating southern and border counties.

How we read the event studies. We therefore rest the interpretation of persistence on $k = 0$ and $k = +1$, which are estimated from all or most cohorts, and treat $k = +2$ and $k = +3$ as suggestive evidence from early adopters rather than as estimates of the average treated county’s longer-run trajectory. The widening confidence intervals at those horizons reflect the same censoring. This does not affect the aggregate ATTs reported throughout the paper, which are averages over the post-treatment observations that exist and are dominated by the cohorts and horizons with the most data.

Table D.1: Cohort Composition by Event Time

Event time	Contributing cohorts	Treated counties	Border or South (%)
$k = +0$	2009, 2010, 2011, 2012	2,980	50.0
$k = +1$	2009, 2010, 2011	1,353	68.7
$k = +2$	2009, 2010	327	91.1
$k = +3$	2009	51	94.1
<i>Cohort composition</i>			
2009 cohort		51	94.1
2010 cohort		276	90.6
2011 cohort		1,026	61.6
2012 cohort		1,627	34.5
Never treated		176	31.2

Notes: The upper panel reports, for each post-treatment event time, the activation cohorts that can contribute given a sample ending in 2012 (cohort g contributes to event time k only if $g + k \leq 2012$), the number of treated counties in those cohorts, and the share of those counties located in border or southern states. The lower panel reports the size and geographic composition of each cohort and of the never-treated group. Border states are Arizona, California, New Mexico, and Texas; southern states follow the Census South region definition. Counties are unweighted in this table.

E Theoretical Framework

This appendix provides an organizing framework for the empirical results. The core mechanism is a negative local labor supply shock where enforcement reduces the effective immigrant workforce through removals, induced out-migration, and reduced inflows. The main propagation channel is hiring frictions. When the local labor pool contracts, recruiting becomes more costly, and firms that depend on continuous hiring are disproportionately affected. A decline in local consumption demand may reinforce this effect. A direct entrepreneurship response is also theoretically possible, although the incorporated self-employment and nonemployer estimates do not support it as the principal empirical channel. The model is partial-equilibrium with labor-only production and is not intended as a structural model to be estimated.

Environment

Consider a local economy in county c with two sectors, $s \in \{T, N\}$, where T denotes tradable and N denotes non-tradable. Firms produce using labor only:

$$y = a n^\alpha, \quad 0 < \alpha < 1, \quad (8)$$

where a is firm-specific productivity and n is labor input. Potential entrepreneurs are of two types, $j \in \{I, D\}$, where I denotes immigrant and D denotes domestic (native-born). Both types draw productivity from the same distribution $G(a)$ with density $g(a)$.

Two baseline exposure measures govern the strength of the enforcement shock. Let $M_c \in [0, 1]$ denote the baseline immigrant share in county c , and let $\Gamma_s \in [0, 1]$ denote sector s 's baseline dependence on immigrant labor. Enforcement intensity $\mathcal{E}_c \geq 0$ reduces the effective local labor pool by a fraction that is increasing in both M_c and Γ_s .

The key departure from a standard competitive labor market model is a recruiting friction. Hiring requires costly search, and the per-hire recruiting cost depends on the tightness of the local labor market. This friction is the central channel through which the labor supply shock propagates to firm entry and expansion decisions.

Labor Supply and Recruiting Frictions

Let L_s denote the effective labor pool available to sector s . Enforcement reduces L_s by removing immigrant workers:

$$L_s(\mathcal{E}_c) = L_s^0 - \phi_s(\mathcal{E}_c, M_c, \Gamma_s), \quad \phi_s > 0, \quad \frac{\partial \phi_s}{\partial \mathcal{E}_c} > 0, \quad (9)$$

where L_s^0 is the pre-enforcement labor pool and ϕ_s captures the reduction in available workers, which is increasing in enforcement intensity and in baseline exposure (M_c, Γ_s) .

Firms that wish to hire n workers incur a recruiting cost:

$$R(n, L_s) = \frac{\kappa}{2} \cdot \frac{n^2}{L_s}, \quad (10)$$

where $\kappa > 0$ governs the severity of recruiting frictions. The cost is increasing and convex in n (more hires are progressively harder to make) and decreasing in L_s (a larger labor pool makes each hire easier). This reduced-form specification captures the idea that recruiting is a bilateral process: when fewer workers are available, each vacancy takes longer to fill and the cost per hire rises. The quadratic form ensures that firms with larger hiring needs are disproportionately affected²²

The key implication is that enforcement raises recruiting costs through a compositional effect on the labor pool:

$$\frac{\partial R}{\partial \mathcal{E}_c} = -\frac{\kappa}{2} \cdot \frac{n^2}{L_s^2} \cdot \frac{\partial L_s}{\partial \mathcal{E}_c} > 0. \quad (11)$$

Because $\partial L_s / \partial \mathcal{E}_c < 0$, the recruiting cost rises. The increase is larger when baseline immigrant exposure (M_c, Γ_s) is higher and when the firm's hiring need n is larger.

Wages. Enforcement also raises equilibrium wages. The labor supply contraction shifts the supply curve inward against a stable demand curve, increasing the wage in both sectors. In tradable sectors, where the output price is exogenous, the wage increase is unambiguous. In non-tradable sectors, the wage increase is attenuated if local demand also falls (see the discussion of local demand below), but the net effect on wages remains positive as long as

²²This formulation can be microfounded through a standard matching function framework (Mortensen and Pissarides, 1994; Pissarides, 2000), where the probability of filling a vacancy is decreasing in the vacancy-to-worker ratio. As the labor pool shrinks, the matching rate falls and firms must post more vacancies (at increasing cost) to achieve the same number of hires. We use the reduced form for tractability. See Hopenhayn et al. (2022) and Karahan et al. (2019) for equilibrium models of firm entry and exit in which labor supply growth governs the startup rate without affecting incumbent exit behavior, precisely the asymmetric pattern we document empirically

the supply contraction dominates the demand contraction. We treat this as a prediction of the model; wage outcomes are not examined in this paper.

Impact on profits. A firm with productivity a that must recruit n workers earns flow profit

$$\pi(a, n, w_s, L_s) = p_s \cdot a n^\alpha - w_s n - R(n, L_s), \quad (12)$$

where p_s is the output price. Enforcement reduces profits through two reinforcing channels: higher wages w_s and higher recruiting costs R . The recruiting cost channel is absent from a frictionless model and creates differential effects across firms with different hiring needs.

Entry versus Incumbent Adjustment

The distinction between recruiting costs and wage levels generates an asymmetry between entrants and incumbents that is central to the empirical results.

Entrants face the full recruiting friction. A potential entrant in sector s pays a sunk cost $F_s > 0$, draws productivity a , and must recruit an entire workforce to begin operating. The entry condition requires that the expected present value of profits, net of recruiting costs, covers the sunk cost:

$$V_s(a) - R(n_s^*(a), L_s) \geq F_s, \quad (13)$$

where $V_s(a)$ is the present value of the ongoing profit stream and $n_s^*(a)$ is the optimal workforce size. The entry cutoff $\bar{a}_s$ satisfies this condition with equality. Because entrants must recruit their full workforce from scratch, the recruiting cost R directly raises the entry threshold. Enforcement increases $\bar{a}_s$ through both the wage channel (which reduces V_s) and the recruiting friction (which raises R).

Incumbents adjust on the intensive margin. An incumbent firm has already recruited its workforce and paid the sunk cost F_s . It continues operating if the flow payoff exceeds per-period operating costs $f_s \geq 0$:

$$\pi(a, n, w_s, L_s) - f_s \geq 0. \quad (14)$$

The incumbent's workforce is already in place. It faces recruiting costs only on replacement hires (turnover) and marginal expansion, not on its entire labor force. For a moderate enforcement shock, the wage increase and the recruiting cost on replacement hires reduce

profits but do not push most incumbents below the shutdown threshold. Firms respond by slowing hiring and operating at a smaller scale rather than exiting.

Expanding incumbents face a partial friction. Continuing establishments that wish to grow must recruit additional workers and therefore face the recruiting friction on new hires.

Why the aggregate hiring response loads on incumbents. The per-firm friction is larger for entrants, but the aggregate employment response need not be, because the friction operates on gross hiring flows wherever they arise and incumbents generate most of those flows. An entrant recruits its workforce n once. An incumbent recruits $\tau_s n$ replacement workers every period, and additional workers whenever it expands. Because continuing establishments vastly outnumber entrants in any period, total gross hiring, and therefore total exposure to the recruiting friction, is dominated by incumbents. In the data, continuing establishments account for about 62% of baseline gross job creation but roughly 85% of its decline (Subsection 5.5).

The model is therefore consistent with two facts that might appear to conflict. The entry margin responds sharply in proportional terms, because the entrant bears the friction on its entire workforce. The incumbent margin accounts for most of the aggregate employment response, because incumbents do most of the hiring. Expansion is also the most discretionary component of an incumbent's labor demand: a firm facing higher recruiting costs can postpone growth without severing the matches it already holds. This is why enforcement compresses job creation at continuing establishments while leaving job destruction and exit largely unchanged, and it is the sense in which incumbents are exposed despite having already paid the founding cost.

The resulting pattern is lower entry, lower job creation, especially at continuing establishments, and limited exit or job destruction. This matches the central empirical findings in Subsections 5.3 and 5.5.

The Role of Turnover

The recruiting friction creates a natural link between sector-level turnover rates and sensitivity to enforcement. In sectors with high baseline turnover, firms must continuously recruit replacement workers simply to maintain their current scale. These sectors are exposed to the recruiting friction not only at entry but on an ongoing basis. When the labor pool shrinks, high-turnover firms face a sustained increase in recruiting costs that low-turnover firms avoid.

Let $\tau_s \in [0, 1]$ denote the baseline separation rate in sector s . Each period, an incumbent firm with workforce n must recruit $\tau_s \cdot n$ replacement workers. The per-period recruiting cost for incumbents is

$$R_{\text{inc}}(\tau_s, n, L_s) = \frac{\kappa}{2} \cdot \frac{(\tau_s \cdot n)^2}{L_s}. \quad (15)$$

This cost is increasing in τ_s : high-turnover sectors bear a larger ongoing recruiting burden. Enforcement amplifies this burden because L_s falls, raising the cost of each replacement hire. For entrants, the effect is even larger because they must recruit their entire workforce (n , not $\tau_s \cdot n$).

Turnover is therefore not a separate mechanism but a channel that amplifies the labor supply shock. The model predicts larger declines in the number of firms, the number of establishments, and entry when τ_s is high, consistent with the descriptive gradient in Table 4.

Exposure Heterogeneity

The labor supply shock and its propagation through recruiting frictions both depend on baseline immigrant exposure. The labor pool contraction ϕ_s is increasing in the county immigrant share M_c and the sector immigrant dependence Γ_s . The recruiting cost increase $\partial R / \partial \mathcal{E}_c$ is therefore also increasing in these exposure measures.

The model predicts that enforcement effects on business counts and entry are stronger when:

- The county immigrant share M_c is higher (larger labor supply shock);
- The sector immigrant labor dependence Γ_s is higher (greater exposure to the shock);
- The sector turnover rate τ_s is higher (greater dependence on continuous hiring);
- These exposure measures interact: high- M_c counties with high- Γ_s sectors experience the largest effects.

Complementary Channel: Immigrant Entrepreneurship

Enforcement could reduce business formation directly by removing or deterring immigrant entrepreneurs. Each period, an immigrant entrepreneur faces probability $\rho_I(\mathcal{E}_c, M_c) \in [0, 1]$ that enforcement forces cessation of operations through detention, deportation, or induced departure. Native entrepreneurs face no direct enforcement risk: $\rho_D = 0$. This reduces the expected present value of an immigrant-owned firm:

$$V_s^I(a) = \frac{\pi_s(a)}{1 - \beta(1 - \rho_I)} < \frac{\pi_s(a)}{1 - \beta} = V_s^D(a), \quad (16)$$

where $\beta \in (0, 1)$ is the discount factor. Immigrant entrepreneurs require higher productivity to justify entry. Because immigrants start firms at higher rates than natives (Kerr and Kerr, 2020; Fairlie and Lofstrom, 2015), and emigration shocks reduce firm creation in origin communities (Anelli et al., 2023), this is a plausible theoretical channel.

The data do not show the predicted entrepreneurship response. The incorporated self-employment estimates for non-citizens, citizens, and all workers are statistically indistinguishable from zero, as is the Census nonemployer estimate (Table 8). The channel may operate for some individuals, but the aggregate evidence does not indicate that it accounts for the decline in employer establishments. The core labor-supply predictions of the model do not depend on it.

Complementary Channel: Local Demand

If enforcement causes immigrant households to leave a county, local consumer spending may fall. This would reduce revenue for non-tradable firms that depend on local demand, while leaving tradable firms (which sell to broader markets) largely unaffected (Moretti, 2010).

If enforcement also reduces local spending, nontradable firms face a compounding effect: higher labor costs from the supply contraction and potentially lower revenue from reduced local demand. This reinforcement is consistent with the negative firm and establishment count estimates in nontradable sectors and their absence in tradable sectors (Table 5).

Two points of clarification are important. First, declining entry, declining job creation, near-zero exit, concentration in immigrant-exposed settings, follow entirely from the labor supply and recruiting friction channels. The demand channel is not necessary for these results; it may strengthen them in non-tradable sectors. Second, the demand channel does not reverse the wage prediction. As long as the labor supply contraction is the dominant force, wages rise in both tradable and non-tradable sectors.

Capital-Labor Substitution

Production in equation (8) uses labor only, so a contraction in the available workforce translates directly into fewer hires and less entry. Firms facing a more expensive workforce may instead substitute toward capital, through equipment, automation, or reorganization that economizes on labor. Because the model excludes that margin, its comparative statics describe the response of a firm that cannot substitute, and they therefore represent an upper bound on how much a given labor supply contraction should suppress hiring.

Three considerations bound how much this matters here. First, the direction of the bias is known and does not threaten identification. If firms do substitute toward capital,

the observed declines in entry and job creation are *smaller* than a labor-only model would predict, and the observed wage response is likewise muted. Substitution therefore attenuates our estimates toward zero rather than generating them: it affects how the magnitudes should be compared with the model, not whether enforcement caused the decline.

Second, the window is short. Our sample covers activations between 2009 and 2012 and observes each county for at most a few post-treatment years. Capital adjustment involves lumpy, largely irreversible investment with well-documented adjustment costs and lags, so the scope for substitution within this horizon is limited even if it would be substantial over a longer one.

Third, the firms driving our results are the least able to substitute. The declines in business counts and entry are concentrated among small establishments, with entry falling 10.3% in the smallest size category while the estimates for medium and large establishments are statistically indistinguishable from zero, and among young establishments (Appendix G). Small and young firms are precisely those for which the capital investment required to economize on labor is hardest to finance. A substitution channel operating mainly through large, well-capitalized firms is consistent with our finding no effect for that group.

We note the limits of this argument. The size and age gradient is also consistent with small and young firms simply being more exposed to immigrant labor, and our data cannot separate the two explanations. A direct test would split sectors by capital intensity, which requires external capital-to-labor measures by industry that we do not use. We therefore treat the labor-only assumption as a simplification whose direction of bias is understood rather than as a feature the data can verify.

F County-Year Aggregation

The main regressions in the paper are estimated at the county-sector-year level, whereas the first stage in Subsection 5.1 is measured at the county-year level, because ACS population counts are not sector-specific. To place the two on the same footing, Table F.1 re-estimates the four business outcomes with the data collapsed to county-year, summing each outcome across sectors within a county and year. The estimates are essentially unchanged; the number of firms falls by 3.9% (versus 3.4% at the sector level), the number of establishments by 3.2% (versus 2.9%), and establishment entry by 8.4% (versus 9.5%), each significant at conventional levels; establishment exit remains statistically indistinguishable from zero. The stability of the entry estimate confirms that it is not an artifact of sector-level aggregation.

At matched aggregation, the business effects are consistent with the first stage. The number of firms and establishments falls by less, in percentage terms, than the contraction in the targeted population documented in Subsection 5.1, implying stock elasticities at or below the free-entry benchmark of one. Establishment entry, an annual flow, falls by more than the population stock but by about as much as the targeted group’s employment, consistent with the flow margin adjusting more sharply than the level of the population.

Table F.1: Business Outcomes at County-Year Aggregation

	Firms (1)	Establishments (2)	Estab. Entry (3)	Estab. Exit (4)
<i>Panel A: County-sector-year (main-text specification)</i>				
SC Activation	−45.2*** (16.4) [−3.4%]	−45.7*** (17.1) [−2.9%]	−18.1** (7.7) [−9.5%]	−2.8 (19.6) [−1.3%]
Observations	410,707	410,707	298,867	301,151
<i>Panel B: County-year (collapsed across sectors)</i>				
SC Activation	−945.2*** (340.6) [−3.9%]	−939.6*** (352.3) [−3.2%]	−265.0** (129.3) [−8.4%]	−138.2 (255.4) [−3.8%]
Observations	24,760	24,760	24,760	24,760
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Sun and Abraham (2021) aggregate ATT for the four business outcomes, 2005–2012. Panel A is the main-text specification at the county-sector-year level, with county, year, and sector fixed effects. Panel B collapses each outcome to the county-year level by summing across sectors and omits sector fixed effects, matching the aggregation of the first stage in Subsection 5.1. Percentage changes are relative to pre-treatment treated-county means, aggregated with the cohort weights of the Sun-Abraham estimator within each panel (Appendix P). Both panels use the full control set, are weighted by baseline county population, and cluster standard errors by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G Which Firms: Heterogeneity by Firm Size and Age

This appendix asks which business categories account for the average response. We use the BDS firm-size and firm-age files, which are county-year panels without a sector dimension, and estimate the same four outcomes within each category. The subgroup estimates are descriptive because we do not estimate formal interactions comparing categories.

Table G.1 groups firms as small (1-19 employees), medium (20-99), and large (100 or more). In the small category, the number of firms falls by 4.2%, the number of establishments by 4.1%, and entry by 10.3%. Exit is statistically indistinguishable from zero. The firm, establishment, and entry estimates are statistically indistinguishable from zero in the medium and large categories. One exception is medium-category exit, which falls by 12.7% and is statistically significant. The size split therefore locates the negative count and entry estimates in small firms, but it does not produce a uniformly null exit pattern across all categories.

Table G.2 groups firms as young (0-5 years), mature (6-10 years), and old (11 years or more). Among young firms, the number of firms falls by 738.8, or 8.1%, the number of establishments by 776.5, or 8.4%, and entry by 266.2, or 11.4%; exit is statistically indistinguishable from zero. Among mature firms, the number of firms and establishments falls by 223.6 and 195.0, or 5.6% and 4.6%, while entry and exit are statistically indistinguishable from zero. All four estimates are close to zero and imprecise among old firms.

The negative entry response is therefore concentrated descriptively among small and young firms. The business-count response is broader across age, appearing among both young and mature firms, but remains concentrated among small firms in the size split. These patterns are consistent with entrants and smaller businesses being more exposed to local recruitment conditions, while the medium-category exit result cautions against reducing the heterogeneity evidence to a single margin.

Table G.1: Heterogeneity by Firm Size

	Firms	Establishments	Estab. Entry	Estab. Exit
<i>Panel A: Small firms (1-19 employees)</i>				
ATT	-837.4*** (296.6)	-829.7*** (296.4)	-262.9** (120.1)	-121.0 (220.1)
Pre-treatment mean	19900.0 [-4.2%]	20045.4 [-4.1%]	2562.0 [-10.3%]	3020.8 [-4.0%]
<i>Panel B: Medium firms (20-99 employees)</i>				
ATT	-98.7 (69.9)	-99.7 (66.0)	2.0 (7.8)	-22.0*** (6.8)
Pre-treatment mean	3262.0 [-3.0%]	4328.3 [-2.3%]	203.0 [+1.0%]	173.9 [-12.7%]
<i>Panel C: Large firms (100+ employees)</i>				
ATT	-5.1 (9.0)	-9.9 (43.7)	-2.5 (16.6)	6.1 (41.5)
Pre-treatment mean	994.1 [-0.5%]	4905.4 [-0.2%]	414.8 [-0.6%]	468.5 [+1.3%]
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes

Notes: [Sun and Abraham \(2021\)](#) aggregate ATT for the number of firms, the number of establishments, establishment entry, and establishment exit, estimated separately by BDS firm-size category using 2005-2012 county-year data. Small: 1-19 employees; medium: 20-99; large: 100 or more. The data have no sector dimension. Percentage changes relative to pre-treatment means appear in square brackets. All specifications include county and year fixed effects with the full control set. Weighted by baseline county population. Standard errors clustered at the county level. Separate-sample estimates do not constitute formal tests of cross-group differences. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table G.2: Heterogeneity by Firm Age

	Firms	Establishments	Estab. Entry	Estab. Exit
<i>Panel A: Young firms (0-5 years)</i>				
ATT	-738.8** (328.8)	-776.5** (333.4)	-266.2*** (90.3)	-141.6 (182.2)
Pre-treatment mean	9070.8 [-8.1%]	9241.2 [-8.4%]	2327.0 [-11.4%]	1761.2 [-8.0%]
<i>Panel B: Mature firms (6-10 years)</i>				
ATT	-223.6** (89.0)	-195.0** (88.6)	-2.8 (21.9)	-4.8 (29.2)
Pre-treatment mean	4021.7 [-5.6%]	4210.5 [-4.6%]	166.8 [-1.7%]	555.7 [-0.9%]
<i>Panel C: Old firms (11+ years)</i>				
ATT	19.9 (77.9)	32.1 (100.8)	1.3 (33.1)	8.7 (55.8)
Pre-treatment mean	11047.0 [+0.2%]	15825.5 [+0.2%]	675.4 [+0.2%]	1333.8 [+0.7%]
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes

Notes: Sun and Abraham (2021) aggregate ATT for the number of firms, the number of establishments, establishment entry, and establishment exit, estimated separately by BDS firm-age category using 2005-2012 county-year data. Young: 0-5 years; mature: 6-10 years; old: 11 years or more, including left-censored firms. The data have no sector dimension. Percentage changes relative to pre-treatment means appear in square brackets. All specifications include county and year fixed effects with the full control set. Weighted by baseline county population. Standard errors clustered at the county level. Separate-sample estimates do not constitute formal tests of cross-group differences. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H Region- and Division-by-Year Fixed Effects

Because SC activated earlier in southern and border counties, a concern is that the design captures differential regional business-cycle dynamics rather than enforcement. If the Great Recession and the recovery affected the South differently through housing, construction, or trade exposure, the Bartik index and the state-level housing interaction we already include may not fully absorb it. This appendix adds fixed effects that do: region-by-year and Census division-by-year. These absorb any shock common to a region in a given year, so the effect is identified only from differences in activation timing *within* a region-year.

This test is informative only if activation timing still varies within region-years. We do not use state-by-year fixed effects precisely because several states activated SC statewide at once, which would absorb the variation entirely (East et al., 2023). Divisions are coarser, and the variation does survive. Each of the nine divisions contains between three and five distinct activation cohorts, and 31 of the 72 division-year cells contain both treated and untreated counties. The comparison is therefore made within region-years rather than across them.

Table H.1 reports the results. Adding region-by-year fixed effects, the number of firms falls by 3.1% against a baseline of 3.4%, establishments by 2.7% against 2.9%, entry by 9.2% against 9.5%, and job creation by 10.2% against 10.3%. Division-by-year fixed effects give 3.3%, 2.9%, 10.1%, and 9.9%. Exit remains statistically indistinguishable from zero in every specification. The estimates are not merely stable but somewhat more precisely estimated: the standard error on the number of firms falls from 16.4 to 13.5, and entry moves from significance at the 5% level to the 1% level.

We read this as evidence against the regional business-cycle confound. If the results were driven by the South and the border recovering differently, absorbing all region-year and division-year shocks should have attenuated them substantially. It does not.

Table H.1: Robustness: Region- and Division-by-Year Fixed Effects

	Firms (1)	Establishments (2)	Estab. Entry (3)	Estab. Exit (4)	Job Creation (5)
<i>Baseline</i>					
SC Activation	−45.2*** (16.4) [−3.4%]	−45.7*** (17.1) [−2.9%]	−18.1** (7.7) [−9.5%]	−2.8 (19.6) [−1.3%]	−394.4** (186.5) [−10.3%]
<i>Region × year</i>					
SC Activation	−41.0*** (13.6) [−3.1%]	−42.2*** (14.1) [−2.7%]	−17.6** (6.8) [−9.2%]	4.9 (16.0) [+2.2%]	−392.4** (181.7) [−10.2%]
<i>Division × year</i>					
SC Activation	−43.5*** (13.5) [−3.3%]	−45.8*** (14.0) [−2.9%]	−19.2*** (6.5) [−10.1%]	4.2 (15.4) [+1.9%]	−380.3** (177.8) [−9.9%]
County Fixed Effects	Yes	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: [Sun and Abraham \(2021\)](#) aggregate ATTs. The baseline includes county, year, and sector fixed effects. The second and third specifications replace the year fixed effect with region-by-year and Census division-by-year fixed effects respectively, absorbing any shock common to a region or division in a given year, so that identification comes only from differences in activation timing within a region-year. Regions are the four Census regions; divisions are the nine Census divisions. Percentage changes relative to pre-treatment treated-county means appear in brackets. All specifications include the full control set, are weighted by baseline county population, and cluster standard errors by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I Selection on Unobservables

A persistent concern in panel identification is that unobserved county-level trends correlated with SC timing could drive the results. We assess sensitivity to selection on unobservables using [Oster \(2019\)](#)’s δ statistic under two values of $R^2_{\max}$: $1.3 \times R^2_{\text{controlled}}$ and $2.0 \times R^2_{\text{controlled}}$.

Table [I.1](#) applies the calculation to all four business outcomes. Adding controls makes the firm and establishment coefficients more negative, moves the entry coefficient toward zero, and moves the exit coefficient close to zero. The implied δ values for firms and establishments are extremely large and positive, while those for entry are extremely large and negative. The exit ratios are -6.3 and -20.8 . In every case, however, the fixed-effects R^2 changes by less than one tenth of a percentage point when controls enter. Because the denominator of the Oster calculation is nearly zero, the ratios are highly unstable. We report them as a directional sensitivity exercise and do not interpret their extreme magnitudes literally.

Table I.1: Oster (2019) Bounds

Outcome	$\hat{\beta}^{\text{naive}}$	$\hat{\beta}^{\text{controlled}}$	R^2_{naive}	$R^2_{\text{controlled}}$	δ (1.3 $\times$)	δ (2.0 $\times$)
Firms	-26.113	-30.180	0.6514	0.6514	95235.574	317451.914
Establishments	-22.362	-30.118	0.6613	0.6613	55276.804	184256.012
Estab. Entry	-22.986	-16.973	0.6724	0.6732	-732.208	-2440.693
Estab. Exit	5.308	0.108	0.6764	0.6771	-6.252	-20.841

Notes: [Oster \(2019\)](#) diagnostics for the number of firms, the number of establishments, establishment entry, and establishment exit. The naive model includes county, year, and sector fixed effects; the controlled model adds the full control set. δ is the proportional selection ratio at the indicated multiple of $R^2_{\text{controlled}}$. The ratios are numerically unstable when adding controls produces almost no change in R^2 , as it does here, and should not be interpreted literally.

J Conley Spatial Standard Errors

County-level outcomes may exhibit positive spatial autocorrelation: neighboring counties share common local economic shocks, which propagate through regional supply chains and housing markets in spatially clustered patterns. Standard errors clustered at the county level account for serial correlation within counties but not for cross-county spatial correlation. We address this using the spatial HAC estimator of [Conley \(1999\)](#), which allows arbitrary correlation across county pairs within a specified geographic cutoff.

Table [J.1](#) reports static TWFE estimates under county clustering and Conley standard errors at 100, 200, and 300 km cutoffs, using county centroids from the Census Gazetteer file. Entry remains significant at the 5% level under every cutoff. The number of firms and establishments remains significant at the 10% level, compared with 5% under county clustering. Exit remains statistically indistinguishable from zero. Spatial adjustment therefore weakens the precision of the count estimates modestly without changing any point estimate.

Table J.1: Robustness to Conley Spatial Standard Errors

Variance estimator	Firms (1)	Establishments (2)	Estab. Entry (3)	Estab. Exit (4)
County clustering	−30.18** (15.16)	−30.12** (14.71)	−16.97** (7.13)	0.11 (3.97)
Conley 100 km	−30.18* (16.67)	−30.12* (16.37)	−16.97** (7.58)	0.11 (4.35)
Conley 200 km	−30.18* (16.16)	−30.12* (15.91)	−16.97** (7.61)	0.11 (4.03)
Conley 300 km	−30.18* (16.04)	−30.12* (15.76)	−16.97** (7.89)	0.11 (3.71)
Pre-treatment mean	1023.0	1217.0	146.7	168.4
Observations	410,707	410,707	298,867	301,151
R^2	0.651	0.661	0.673	0.677
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes

Notes: Each row reports the same static TWFE point estimates with a different variance estimator. These point estimates are not the interaction-weighted ATTs in the main tables. Conley standard errors use spherical distances between county centroids (Census 2020 Gazetteer) with the indicated radius cutoff ([Conley, 1999](#)). All specifications include county, year, and sector fixed effects with the full control set. Weighted by baseline county population. Estimation sample: 2005-2012. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

K Spatial Unit Root Diagnostics

Müller and Watson (2024) show that strong spatial persistence, the spatial analogue of a unit root in time series, can produce spurious regression results even when standard spatial HAC corrections are applied. If economic variables exhibit very slow spatial decay in dependence, Conley standard errors underestimate the true variance, and t -statistics may be inflated by orders of magnitude. This concern is most acute for cross-sectional persistence regressions, but any empirical design using spatially distributed data should assess whether the findings survive corrections that account for this possibility.

To test whether our results reflect spurious spatial correlation, we conduct placebo tests following the approach of Conley and Kelly (2025). We construct 1,000 spatially correlated placebo treatments by drawing county-level noise from an exponential covariance field with a 200 km spatial range, matching the distance cutoff from the Conley standard errors in Appendix J. Each placebo preserves the real SC cohort structure: the same number of counties activate in each year, but *which* counties are assigned to each cohort is determined by the spatially correlated noise. This ensures the placebo treatments have both the temporal variation needed to survive county fixed effects and spatial correlation comparable to the real treatment.

Table K.1 reports the results. In the county-year panel, the real coefficients are -27.85 for the number of firms, -27.41 for the number of establishments, -16.25 for entry, and 1.04 for exit. The coefficient-rank p -values are 0.039, 0.033, 0.058, and 0.872, respectively. Using t -statistics, which incorporate the estimate-specific standard errors, the corresponding rank p -values are 0.109, 0.102, 0.046, and 0.874. Entry is therefore in the tail of the placebo distribution under the t -statistic comparison; the business counts are unusual but do not cross conventional thresholds under that comparison, and exit lies near the center.

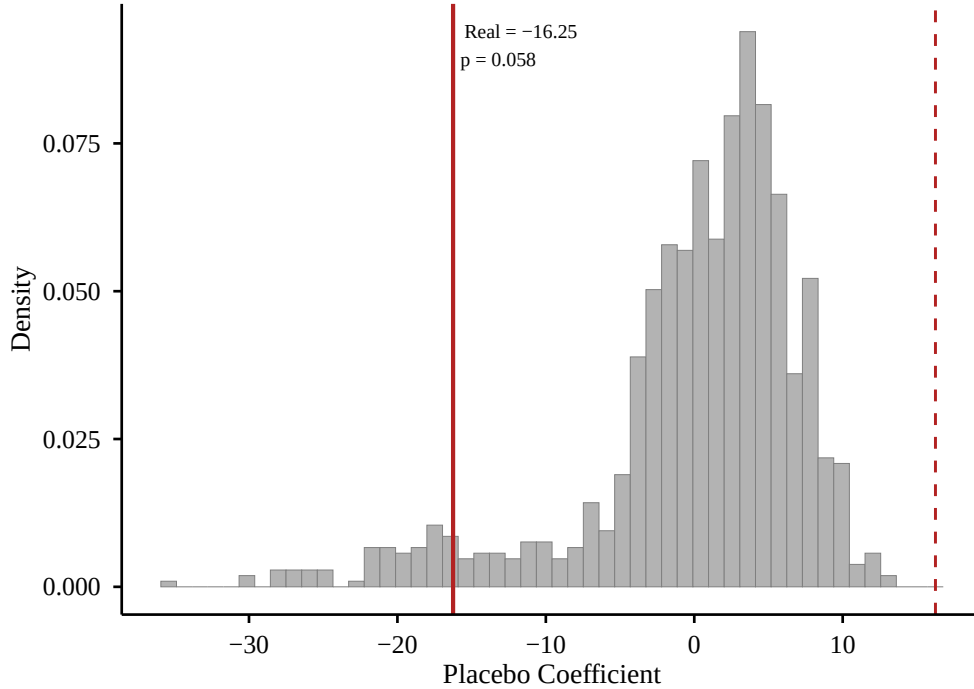
Panel B reports county-clustered and Conley 95% confidence intervals. The county-clustered intervals exclude zero for the number of firms, the number of establishments, and entry. At the 200 km Conley cutoff, the count intervals include zero while the entry interval remains negative. Figure K.1 plots the placebo distribution for entry.

Our panel difference-in-differences design is structurally less exposed to the spatial unit root problem than the cross-sectional persistence regressions that motivate Müller and Watson (2024). Identification comes from within-county temporal variation in enforcement timing, not from cross-sectional level differences in spatially persistent variables. County fixed effects absorb the cross-sectional spatial levels that are most vulnerable to spatial $I(1)$ behavior, and the staggered rollout of Secure Communities provides temporal variation that is orthogonal to spatial persistence by construction.

Table K.1: Spatial Unit Root Diagnostics

	Number of Firms (1)	Number of Establishments (2)	Establishment Entry (3)	Establishment Exit (4)
<i>Panel A: Spatial Noise Placebo Tests (1,000 sims)</i>				
Real estimate	-27.850	-27.407	-16.250	1.043
Placebo mean (std. dev.)	0.303 (12.233)	0.243 (11.049)	0.149 (7.427)	0.145 (5.494)
Rank p -value (coef.)	0.039	0.033	0.058	0.872
Rank p -value (t -stat)	0.109	0.102	0.046	0.874
Quantile position	0.039	0.033	0.058	0.525
<i>Panel B: 95% Confidence Intervals</i>				
County clustered	[-59.885, -0.474]	[-58.959, -1.277]	[-30.947, -3.000]	[-7.678, 7.894]
Conley 200 km	[-61.859, 1.500]	[-61.301, 1.065]	[-31.880, -2.067]	[-7.796, 8.012]
Spatial placebo [2.5, 97.5%]	[-30.815, 17.582]	[-28.677, 16.609]	[-20.790, 9.747]	[-9.223, 12.724]

Notes: Panel A reports results from 1,000 spatial noise placebo simulations following [Conley and Kelly \(2025\)](#). Each placebo assigns spatially correlated noise (exponential covariance, 200 km range) to determine which counties are treated, preserving the real SC cohort-size distribution and activation timing. The rank p -value is the share of placebo $|t|$ -statistics exceeding the real value (two-sided). Panel B reports county-clustered 95% confidence intervals. All specifications use county and year fixed effects with the full control set on a county-year panel (collapsed across sectors). Weighted by baseline county population.

**Figure K.1:** Spatial Noise Placebo Distribution: Establishment Entry

Notes: Distribution of 1,000 placebo coefficients from regressions of establishment entry on spatially correlated noise treatments. Each placebo preserves the real SC cohort structure but reassigns treated counties using spatially correlated noise with an exponential covariance and 200 km range. The solid red line marks the real SC estimate (-16.25); the dashed red line marks its symmetric positive value. The real coefficient is at the 5.8th percentile of the placebo coefficient distribution, and the two-sided rank p -value based on t -statistics is 0.046.

L Spatial Spillovers and Displacement

Table L.1 reports the own-SC ATT for the four business dynamics outcomes as spillover-exposure controls enter one at a time, following the difference-in-differences spillover framework of Butts (2024) (see also Clarke (2017)). Exposure is the population-weighted Secure Communities coverage of (i) other counties in the county’s own commuting zone and (ii) counties in other commuting zones within 100 miles, entering for control county-years only.

Table L.1: Own-SC Effects Under Spillover-Exposure Controls

	(1) No exposure	(2) Within-CZ only	(3) Cross-CZ only	(4) Both
<i>Panel: Number of Firms</i>				
Own SC (ATT)	-45.2*** (16.4)	-61.4*** (20.5)	-41.3*** (15.8)	-51.6*** (18.3)
Within-CZ exposure		-39.3* (21.5)		-43.2** (22.0)
Cross-CZ exposure			6.4 (4.3)	18.5*** (6.9)
<i>Panel: Number of Establishments</i>				
Own SC (ATT)	-45.7*** (17.1)	-62.2*** (20.6)	-42.4** (16.5)	-52.9*** (18.6)
Within-CZ exposure		-40.1* (21.4)		-43.8** (22.0)
Cross-CZ exposure			5.4 (5.1)	17.7** (7.6)
<i>Panel: Establishment Entry</i>				
Own SC (ATT)	-18.1** (7.7)	-22.0** (10.6)	-12.2 (13.4)	-15.3 (14.5)
Within-CZ exposure		-9.5 (9.8)		-12.1 (9.4)
Cross-CZ exposure			9.5 (12.8)	12.7 (13.0)
<i>Panel: Establishment Exit</i>				
Own SC (ATT)	-2.8 (19.6)	-6.4 (20.7)	-9.3 (23.4)	-11.0 (23.9)
Within-CZ exposure		-8.5 (6.0)		-6.7 (4.3)
Cross-CZ exposure			-10.5 (14.2)	-8.7 (13.6)

Notes: [Sun and Abraham \(2021\)](#) aggregate ATTs for the number of firms, the number of establishments, establishment entry, and establishment exit. All columns use the identical exposure-universe sample; column (1) reproduces the baseline. Exposure variables are linear population-weighted SC coverage shares of neighboring counties within the same commuting zone and in other commuting zones within 100 miles of the county centroid, entering for untreated county-years only ([Butts, 2024](#)). The correlation between the two exposure shares on control county-years is 0.55. The number of untreated and unexposed counties falls from 3,096 in 2005-2008 to 245 in 2011 and zero in 2012. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

M Alternative Clustering

Table M.1 reports static TWFE estimates under county clustering, state clustering, and two-way clustering by county and year. Point estimates are identical across rows by construction. The number of firms and establishments is significant at 5% under county clustering, at 10% under state clustering, and statistically indistinguishable from zero under two-way clustering. Entry is significant at 5% under county and state clustering and at 10% under two-way clustering. Exit remains statistically indistinguishable from zero.

Table M.1: Robustness to Alternative Clustering

	Firms (1)	Establishments (2)	Estab. Entry (3)	Estab. Exit (4)
County clustering	−30.180** (15.156)	−30.118** (14.715)	−16.973** (7.129)	0.108 (3.972)
State clustering	−30.180* (16.752)	−30.118* (17.446)	−16.973** (7.274)	0.108 (3.405)
Two-way (county and year)	−30.180 (17.750)	−30.118 (18.591)	−16.973* (8.242)	0.108 (3.015)
Observations	410,707	410,707	298,867	301,151
R^2	0.651	0.661	0.673	0.677
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Each row reports the same static TWFE point estimates with a different variance estimator; these are not the interaction-weighted ATTs in the main tables. All specifications include county, year, and sector fixed effects with the full control set. Weighted by baseline county population. Estimation sample: 2005-2012. Where required, `fixest` applies its positive-semidefinite correction to the two-way covariance matrix. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

N Unweighted Estimates

Table N.1 compares weighted and unweighted static TWFE estimates. Without population weights, the number of firms, the number of establishments, and entry remain negative and statistically significant, while exit remains statistically indistinguishable from zero. The level coefficients are much smaller without weights because the unweighted estimand gives equal influence to small and large counties; they should not be compared mechanically with the weighted magnitudes.

Table N.1: Robustness: Weighted vs. Unweighted Estimates

	Firms	Establishments	Estab. Entry	Estab. Exit
<i>Panel A: Weighted by baseline population</i>				
Secure Communities	-30.180** (15.156)	-30.118** (14.715)	-16.973** (7.129)	0.108 (3.972)
Observations	410,707	410,707	298,867	301,151
R^2	0.651	0.661	0.673	0.677
<i>Panel B: Unweighted</i>				
Secure Communities	-2.744*** (0.626)	-2.430*** (0.663)	-2.399*** (0.394)	0.051 (0.302)
Observations	410,707	410,707	298,867	301,151
R^2	0.599	0.609	0.617	0.624
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Panel A weights by baseline county population (2000 Census); Panel B is unweighted. Both panels report static TWFE estimates with the full control set and county, year, and sector fixed effects. Weighting changes the estimand, so level coefficients are not directly comparable across panels. Standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

O Leave-One-Cohort-Out Stability

Table O.1 re-estimates the Sun-Abraham aggregate ATT after dropping each activation cohort. The number of firms and establishments remains negative and statistically significant in every subsample. Entry remains negative, with ATTs from -17.7 to -21.7 and significance levels from 10% to 1%. Exit remains statistically indistinguishable from zero. No single activation cohort accounts for the negative count or entry estimates.

Table O.1: Leave-One-Cohort-Out Robustness

	Number of Firms	Number of Establishments	Establishment Entry	Establishment Exit
Full sample	-45.2*** (16.4)	-45.7*** (17.1)	-18.1** (7.7)	-2.8 (19.6)
Drop 2009	-45.6*** (16.3)	-45.6*** (16.1)	-17.7* (9.7)	-1.6 (21.0)
Drop 2010	-51.2*** (16.2)	-53.5*** (16.9)	-21.7*** (6.4)	1.9 (10.4)
Drop 2011	-46.2** (21.5)	-46.0** (22.8)	-18.7** (9.3)	-11.0 (21.7)
Drop 2012	-47.6** (19.2)	-48.9** (20.4)	-17.7** (7.2)	-2.7 (22.9)
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Sector Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Each row drops the indicated SC activation cohort and re-estimates the [Sun and Abraham \(2021\)](#) aggregate ATT on the remaining sample. All specifications include county, year, and sector fixed effects, weighted by baseline county population. Standard errors clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

P Sensitivity to the Percentage Normalization

Every percentage effect reported in the paper divides a level ATT by a pre-treatment baseline. This appendix reports how those percentages move under alternative constructions of that baseline. The level estimates, standard errors, and inference are identical throughout; only the denominator changes.

The [Sun and Abraham \(2021\)](#) aggregate ATT is a weighted average of cohort-by-period treatment effects, with weights given by each cohort’s share of population-weighted post-treatment cells. Those weights are 0.175, 0.406, 0.294, and 0.125 for the 2009 through 2012 cohorts. Baseline establishment entry per county-sector cell is 302.2 for the 2009 cohort and 284.9 for 2010, against 52.9 and 52.5 for 2011 and 2012, so early cohorts are roughly six times larger at baseline and carry more than half the ATT weight. A baseline that pools all treated counties equally therefore describes a different population from the one the estimand weights, and understates the denominator.

Column (3) of Table [P.1](#) aggregates the cohort-specific baselines with these weights and is the construction used throughout the paper. Column (2) reports the naive construction that pools all treated counties at their own pre-activation year without applying cohort weights. Because each cohort’s baseline is measured at its own year before activation, the pooled and cohort-weighted columns both draw on different calendar years across cohorts: the 2009 cohort from 2008, the 2012 cohort from 2011. Entry among not-yet-treated counties fell from 193.0 per cell in 2007 to 84.5 in 2010, so this matters for the flow outcomes. Columns (4) and (5) address it by replacing the single pre-treatment year with the three-year pre-treatment average and with a fixed 2005-2007 window common to every cohort.

The stock outcomes are unaffected by any of these choices, since stock levels moved little over the sample period. The flow outcomes vary: establishment entry ranges from -7.8% to -9.5% and job creation from -8.2% to -10.3% . We report column (3) in the paper, which is both the conventional single-year construction and the largest in magnitude of the three corrected columns.

Table P.1: Percentage Effects under Alternative Baseline Denominators

	Level ATT (1)	Percent of baseline under alternative denominators			
		Published (2)	Cohort-weighted (3)	Three-year (4)	Fixed 2005-07 (5)
Number of firms	−45.2*** (16.4)	−4.4%	−3.4%	−3.4%	−3.4%
Number of establishments	−45.7*** (17.1)	−3.8%	−2.9%	−2.9%	−2.9%
Establishment entry	−18.1** (7.7)	−12.3%	−9.5%	−8.6%	−7.8%
Establishment exit	−2.8 (19.6)	−1.7%	−1.3%	−1.3%	−1.5%
Job creation	−394.4** (186.5)	−13.8%	−10.3%	−9.0%	−8.2%
Job destruction	109.7 (594.0)	+3.0%	+2.2%	+2.3%	+2.5%

Notes: Column (1) reports the [Sun and Abraham \(2021\)](#) aggregate ATT in levels, with standard errors clustered by county in parentheses; it is identical across the remaining columns. Columns (2) to (5) express that same estimate as a percentage of a pre-treatment baseline, varying only how the baseline is constructed. Column (2) pools all treated counties at their own year before activation. Column (3) aggregates cohort-specific baselines using the cohort weights of the ATT itself and is the construction reported throughout the paper. Column (4) replaces the single pre-treatment year with the three-year pre-treatment average, and column (5) with a fixed 2005-2007 window common to every cohort; both retain the ATT cohort weights. All specifications include county, year, and sector fixed effects with the full control set, are weighted by baseline county population, and cluster standard errors by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.